

A continuum and computational framework for viscoelastodynamics: III. A nonlinear theory

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ABSTRACT

We continue our investigation of viscoelasticity by extending the Holzapfel–Simo approach discussed in Part I to the fully nonlinear regime. By scrutinizing the relaxation property for the non-equilibrium stresses, it is revealed that a kinematic assumption akin to the Green–Naghdi type is necessary in the design of the potential. This insight underscores a link between the so-called additive plasticity and the viscoelasticity model under consideration, further inspiring our development of a nonlinear viscoelasticity theory. Our strategy is based on Hill's hyperelasticity framework and leverages the concept of generalized strains. Notably, the adopted kinematic assumption makes the proposed theory fundamentally different from the existing models rooted in the notion of the intermediate configuration. The computation aspects, including the constitutive integration, consistent linearization, and modular implementation, are addressed in detail. A suite of numerical examples is provided to demonstrate the capability of the proposed model in characterizing viscoelastic material behaviors at large strains.

1. Introduction

1.1. Motivation

The motivation of this work is to generalize the finite deformation *linear* viscoelasticity models to the nonlinear regime, and it came from observations of our previous studies. Throughout this work, we refer to Liu, Latorre, and Marsden [1] as Part I, and it was revealed in Part I that the finite deformation linear viscoelasticity models, under the isothermal condition, are characterized by the following form of the configurational free energy

$$Y(\tilde{\mathbf{C}}, \mathbf{I}) = \frac{1}{4\mu} \left| 2 \frac{\partial \mathcal{G}(\tilde{\mathbf{C}})}{\partial \tilde{\mathbf{C}}} - \mu (\mathbf{I} - \mathbf{I}) - \hat{\mathbf{S}}_0 \right|^2. \quad (1.1)$$

The constant symmetric tensor $\hat{\mathbf{S}}_0$ determines the initial value of $\mathbf{Q} := -2\partial Y/\partial \mathbf{I}$. The linearity of such type of models comes from the above quadratic form of the energy Y and the Newtonian assumption on the viscous behavior (i.e., there exists a constant rank-four viscosity tensor relating the stress and strain rate). The models based on the identical-polymer-chain assumption are recovered by taking the energy-like function $\mathcal{G}$ in (1.1) as the hyperelastic free energy scaled by a non-dimensional parameter. Although such type of modeling approach was successfully adopted in a variety of scenarios [2–5], its configurational free energy lacks a clear physical interpretation. Another model is coined as the Holzapfel–Simo–Saint Venant–Kirchhoff model in Part I, which is instantiated

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Table 1

Elastoplastic and viscoelasticity models based on two kinematic assumptions. The viscoelasticity model based on the additive kinematic assumption (i.e., the lower right corner of the table) is also referred to as the finite deformation linear viscoelasticity [1], meaning its nonlinearity is only geometrical.

	Multiplicative decomposition	Additive decomposition
Elastoplasticity	Lee 1969 [13]	Green & Naghdi 1965 [6], Miehe 1998 [14]
Viscoelasticity	Sidoroff 1974 [15], Reese & Govindjee 1998 [16]	Simo 1987 [10], Holzapfel & Simo 1996 [3]

by taking

$$\mathcal{G} = \mu \left| \frac{\tilde{\mathbf{C}} - \mathbf{I}}{2} \right|^2,$$

and $\hat{\mathbf{S}}_0 = \mathbf{O}$. That leads to a simpler form of the configurational free energy, i.e.,

$$\Upsilon(\tilde{\mathbf{C}}, \mathbf{F}) = \mu \left| \frac{\tilde{\mathbf{C}} - \mathbf{F}}{2} \right|^2. \quad (1.2)$$

The significance of the above energy can be made clearer if we introduce the volume-preserving Lagrangian strain $\tilde{\mathbf{E}}$ and a viscous counterpart $\mathbf{E}^v$ as

$$\tilde{\mathbf{E}} := \frac{\tilde{\mathbf{C}} - \mathbf{I}}{2}, \quad \text{and} \quad \mathbf{E}^v := \frac{\mathbf{F} - \mathbf{I}}{2}.$$

Then the energy (1.2) can be expressed as

$$\Upsilon(\tilde{\mathbf{C}}, \mathbf{F}) = \mu |\tilde{\mathbf{E}} - \mathbf{E}^v|^2. \quad (1.3)$$

One insight one may glean from the energy form (1.3) is that the two strain-like tensors enter into the energy function in the form of $\tilde{\mathbf{E}} - \mathbf{E}^v$. This closely resembles the kinematic assumption made by Green and Naghdi in the realm of elastoplasticity modeling [6–9]. If one views the tensor $\tilde{\mathbf{E}} - \mathbf{E}^v$ as the strain of the non-equilibrium elastic part, the rationale behind labeling the model “Saint Venant-Kirchhoff” becomes evident. This perhaps is not entirely surprising, considering the original model introduced by Simo in 1987 [10] is a geometrically nonlinear extension of the small-strain viscoelasticity theory, which also relies on an additive decomposition of the strain [11, Chapter 10]. However, adopting the Saint Venant-Kirchhoff model is restrictive in material modeling and may suffer from mathematical issues [12]. In this work, we extend the model to the nonlinear regime by adopting the kinematic decomposition introduced initially by Green and Naghdi in elastoplasticity [6,7] as our point of departure.

1.2. Kinematic assumptions in finite inelasticity

The above discussion brings us to a fundamental issue in inelasticity modeling, that is, how to quantify the inelastic kinematic response. There have been long-lasting debates on this issue, and there is no commonly-accepted approach to date. A widely adopted approach is through the multiplicative decomposition of the deformation gradient $\mathbf{F} = \mathbf{F}^e \mathbf{F}^i$, in which the elastic and inelastic deformations $\mathbf{F}^e$ and $\mathbf{F}^i$ are related to an imaginary stress-free intermediate configuration. Although this approach is nowadays often credited to Lee and his associates for their pioneering work in the late 1960s [13,17], this idea can be traced back to earlier works of Eckart [18] and Kröner [19], among others [20,21]. Mathematically, the plastic deformation gradient belonging to $GL(3)_+$ is used as the internal state variable for characterizing inelastic deformation. Physically, its microscopic interpretation is due to the crystallographic slip in single-crystal metal plasticity, according to the continuum slip theory [22,23]. This strategy has been found to be applicable in thermomechanics [24], growth mechanics [25], etc. In a broader context, this modeling framework is mature, with computational approaches and engineering applications extensively investigated [11,26,27]. The adoption of the multiplicative decomposition for viscoelasticity modeling can be traced back to the work of Sidoroff in 1974 [15]. Building upon this work, Reese and Govindjee systematically constructed a viscoelasticity model and thoroughly discussed the corresponding finite element modeling [16,28]. In particular, their algorithmic treatment of the constitutive integration exploited the techniques in computational plasticity [29–32]. That viscoelasticity model, derived from thermomechanical principles, allows the use of general nonlinear potential to address the non-equilibrium aspects. Those attributes make it appealing, and thus the multiplicative viscoelasticity model has since then been generalized to modeling additional material behaviors, especially for soft materials. Representative applications include electroactive polymers [33,34], anisotropic materials [35–38], shape memory polymers [39], and hydrogels [40], to list a few.

Nevertheless, the concept of the intermediate configuration is not without shortcomings, and its validity has been questioned since its inception. A primary concern comes with the existence and uniqueness of the intermediate configuration [7,41,42]. Moreover, the intermediate configuration can be arbitrarily rotated without affecting the multiplicative decomposition. Therefore, the theory is incomplete unless the description of the rotational parts of $\mathbf{F}^e$ and $\mathbf{F}^i$ are specified. Furthermore, the objectivity of the constitutive models essentially poses a rather involved constraint governing $\mathbf{F}^e$ and $\mathbf{F}^i$, which is sometimes known as the invariance requirements [43–45]. There are several works striving to resolve the aforementioned issues by proposing proper kinematic laws [46,47]. However, they are more or less ad hoc and can be difficult to be justified from physical experiments. In our opinion, those ad hoc laws should be placed together with the multiplicative decomposition as the primary modeling assumptions

for specific models. In anisotropic cases, this is particularly critical as one relies on the specific assumption to define the evolution of the material symmetry groups on the intermediate configurations [35,38,48].

Almost parallel to the development of the theory based on the multiplicative decomposition, Green and Naghdi pioneered an alternative strategy for elastoplasticity based on an additive decomposition [6]. They introduced a symmetric tensor E^p and referred to it as the plastic strain. In their theory, the difference $E - E^p$ enters into the energy function as a tensorial variable. Recognizing the potential incompatibility issue, both authors underscored that $E - E^p$ shall not be interpreted as the elastic strain and thus avoided the terminology “additive decomposition”. Despite that, the interpretation of additive decomposition has been used by others since then [8,9]. An extension of this concept was made by Miehe, who invoked the additive split of the Hencky strain as the point of departure [14,49–51]. Schröder et al. considered anisotropic elastoplasticity within this framework [52]. Papadopoulos and Lu expanded the scope of the additive plasticity theory by exploiting the Seth–Hill strain family [53,54]. Computational aspects of this type of model have also been investigated [55–57]. Notwithstanding the accumulated studies in the additive theory, research efforts have been made predominantly in the realm of elastoplasticity. As discussed earlier in this article, the model of Simo [3,10] implicitly adopted the additive decomposition. The concept of strain-like internal state variables can be traced back even earlier to the molecular theory for polymers by Green and Tobolsky [58]. We summarize key inelasticity models in Table 1 based on their kinematic assumptions. Given that the existing additive viscoelasticity model is only geometrically nonlinear, it is imperative and feasible to systematically extend it to the fully nonlinear regime and devise corresponding computation methodologies.

1.3. Hyperelasticity of Hill's class and generalized strains

In the phenomenological modeling of elastic materials, it is nowadays fairly common to design nonlinear functions written in terms of the principal stretches or invariants to characterize nonlinear hyperelastic behaviors [59,60]. Alternatively, in a different approach proposed by Hill [61], the structure of Hooke's law is maintained, and the nonlinearity is manifested through generalized strains. In [62], Hill proposed the concept of general strains by adopting a smooth monotone scale function of the principal stretches. The first instantiation of this concept is the Seth–Hill [61–63] or Doyle–Ericksen strains [64], which constitute a strain family parameterized by a real number. With this strain family, Hill's hyperelasticity framework involves the Saint Venant–Kirchhoff model [65, pp. 250–251] as well as the Hencky elasticity model [66,67]. This strain family also serves as the backbone for the aforementioned additive plasticity [14,51,53,54,57,68]. Nevertheless, an apparent drawback of the Seth–Hill strain family, except the Hencky strain, is that they are not coercive, meaning that the strains remain finite for extreme deformations [69]. To rectify this issue, coercivity is supplemented as an additional requirement for the scale function of generalized strains. In 1991, Curnier and Rakotomanana proposed a two-integer-parameter strain family that incorporates the strains (e.g. the Pelzer, Mooney, Wall, and Rivlin strains) not belonging to the Seth–Hill family [70]. This strain family was later extended by Darigani and Naghdabadi by allowing the parameters to take real values [71]. Around the end of the twentieth century, Bažant and Itskov each individually designed a single-parameter strain family with motivations from different fields [72,73]. In 2006, Curnier and Zysset proposed a two-parameter strain family that is easy to compute and delivers good approximations of the Seth–Hill family [74]. Later in 2013, Darigani and Naghdabadi introduced a two-parameter strain family represented in the exponential form [75], which can be beneficial for describing the strain hardening in very large deformations.

Furthermore, Hill's modeling strategy has evolved in another aspect. Traditionally, a single pair of work-conjugate stress and strain is utilized, meaning the strain energy involves a single quadratic term of strain. Considering the Ogden model on the other hand, three terms, or six parameters, are needed to achieve satisfactory fitting results for realistic materials [59,65,76]. Notably, the invariants of the Seth–Hill strains resemble the energy form of the Ogden model [77], leading to the introduction of additional quadratic terms of different strains into the potential function for material modeling purposes [71,78]. Over the past several decades, we have witnessed the progressive enrichment of Hill's hyperelasticity and generalized strains. The ever-increasing modeling strategies offer new possibilities in modeling both elastic and inelastic materials.

1.4. Structure and content of the article

The body of this work is organized as follows. In Section 2, the visco-hyperelasticity theory is methodically developed. The emphases are placed on the generalized strains, kinematic assumptions of Green–Naghdi type, constitutive relations based on the Helmholtz and Gibbs potentials. Section 3 offers comprehensive discussions on consistent linearization, constitutive integration, and modular implementation. Section 4 showcases numerical examples that underscore the efficacy of our proposed model in characterizing viscoelastic material properties. We draw conclusions and discuss directions for future studies in Section 5.

2. Theory

In this section, we construct a nonlinear visco-hyperelasticity theory based on the kinematic assumptions of the Green–Naghdi type. The generalized strains are utilized to treat the material nonlinearity within the framework of Hill's hyperelasticity. Within this framework, the existing finite linear viscoelasticity is recovered when the strain is specialized to the Green–Lagrange strain. At the outset of our presentation, the following clarifications are made.

1. The summation convention is *not* adopted, and all summations are expressed explicitly with a summation sign.
2. The norms of a vector $\mathbf{W}$ and a rank-two tensor $\mathbf{A}$ are defined as

$$|\mathbf{W}| := (\mathbf{W} \cdot \mathbf{W})^{\frac{1}{2}} \quad \text{and} \quad |\mathbf{A}| := (\text{tr}[\mathbf{A}\mathbf{A}^T])^{\frac{1}{2}} = (\mathbf{A} : \mathbf{A})^{\frac{1}{2}}.$$

3. The rank-two (rank-four) identity and zero tensors are denoted by $\mathbf{I}$ ($\mathbb{I}$) and $\mathbf{O}$ ($\mathbb{O}$), respectively.

2.1. Kinematics and generalized strains

We adopt the initial configuration of the body $\Omega_X \subset \mathbb{R}^3$ as the reference configuration and assume it to be stress-free. The motion of the body is described by a family of diffeomorphisms $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t) = \boldsymbol{\varphi}_t(\mathbf{X})$ parameterized by the time coordinate t . For a given time coordinate, $\boldsymbol{\varphi}_t$ maps a material point initially at $\mathbf{X} \in \Omega_X$ to the location $\mathbf{x}$. We denote the current configuration of the body as $\Omega_x^t := \boldsymbol{\varphi}(\Omega_X, t)$. The displacement is defined as $\mathbf{U}(\mathbf{X}, t) := \boldsymbol{\varphi}(\mathbf{X}, t) - \mathbf{X}$, and the velocity is defined as $\mathbf{V} := d\mathbf{U}/dt$, where $d(\cdot)/dt$ designates the total time derivative. The deformation gradient and the Jacobian are defined as

$$\mathbf{F} := \frac{\partial \boldsymbol{\varphi}_t}{\partial \mathbf{X}} \quad \text{and} \quad J := \det(\mathbf{F}).$$

To distinguish bulk and shear responses, the deformation gradient is multiplicatively decomposed into volumetric and volume-preserving parts as

$$\mathbf{F} = \mathbf{F}_{\text{vol}} \tilde{\mathbf{F}}, \quad \mathbf{F}_{\text{vol}} := J^{1/3} \mathbf{I}, \quad \tilde{\mathbf{F}} := J^{-1/3} \mathbf{F}. \quad (2.1)$$

The right Cauchy–Green tensor $\mathbf{C} := \mathbf{F}^T \mathbf{F}$ and its unimodular counterpart $\tilde{\mathbf{C}} := \tilde{\mathbf{F}}^T \tilde{\mathbf{F}}$ enjoy the following spectral representations,

$$\mathbf{C} = \sum_{a=1}^3 \lambda_a^2 \mathbf{M}_a \quad \text{and} \quad \tilde{\mathbf{C}} = \sum_{a=1}^3 \tilde{\lambda}_a^2 \mathbf{M}_a, \quad (2.2)$$

in which λ_a ($\tilde{\lambda}_a := J^{-1/3} \lambda_a$) are the (modified) principal stretches, and $\mathbf{M}_a := \mathbf{N}_a \otimes \mathbf{N}_a$ are the self-dyads of the principal referential directions $\mathbf{N}_a$. In particular, the set of self-dyads $\{\mathbf{M}_a\}$ forms the basis for the tensor space of commutators of $\mathbf{C}$. Recall that

$$\mathbf{F} \mathbf{N}_a = \lambda_a \mathbf{n}_a,$$

where the unit-length spatial vectors $\mathbf{n}_a$ are known as the principal spatial directions. With the above, the spectral representations of $\mathbf{F}$ and $\tilde{\mathbf{F}}$ take the form

$$\mathbf{F} = \sum_{a=1}^3 \lambda_a \mathbf{n}_a \otimes \mathbf{N}_a \quad \text{and} \quad \tilde{\mathbf{F}} = \sum_{a=1}^3 \tilde{\lambda}_a \mathbf{n}_a \otimes \mathbf{N}_a.$$

In this study, strain plays a pivotal role in characterizing nonlinear material behaviors, and we consider a general Lagrangian strain represented as

$$\mathbf{E} := \sum_{a=1}^3 E(\lambda_a) \mathbf{M}_a, \quad (2.3)$$

with $E : (0, \infty) \rightarrow \mathbb{R}$ being a scale function. The generalized strain $\tilde{\mathbf{E}}$ associated with the isochoric part of the deformation is analogously defined as

$$\tilde{\mathbf{E}} := \sum_{a=1}^3 E(\tilde{\lambda}_a) \mathbf{M}_a. \quad (2.4)$$

The Eulerian counterparts of the above two Lagrangian strains can be defined by replacing $\mathbf{M}_a$ with $\mathbf{n}_a \otimes \mathbf{n}_a$. It can be verified that the generalized strains defined above are objective and isotropy [65]. It is demanded by Hill [61] that, to make the above two definitions suitable, the scale function E has to be twice differentiable,¹ monotonically increasing (i.e., $E' > 0$), and satisfy

$$E(1) = 0 \quad \text{and} \quad E'(1) = 1. \quad (2.5)$$

The monotonicity is physically reasonable and ensures the inverse of the function E exists. It is also termed as the regularity condition [70]. The condition (2.5)₁ guarantees the measure vanishes for the undeformed state. The condition (2.5)₂ renders the measure compatible with the infinitesimal strain and is known as the normality condition [79]. The well-known Seth–Hill strain family satisfies the above attributes. Nevertheless, it is found that, except for the Hencky strain [20,66], the rest members of the strain family exhibit erratic behaviors when the deformation ratio asymptotically approaches infinity or zero. Therefore, two additional conditions were supplemented, demanding the scale function E to be coercive [69], i.e.,

$$\lim_{\lambda \rightarrow \infty} E(\lambda) \rightarrow +\infty \quad \text{and} \quad \lim_{\lambda \rightarrow 0} E(\lambda) \rightarrow -\infty. \quad (2.6)$$

They require the strain to approach infinity when extreme deformation presents. Furthermore, the strain is tension–compression symmetric if the scale function E satisfies

$$E(\lambda^{-1}) = -E(\lambda). \quad (2.7)$$

While the symmetry condition (2.7) is mathematically elegant [69,72], it is important to note that materials typically behave asymmetrically in tension and compression [77,80,81]. In this regard, certain two-parameter generalized strains enable one to flexibly control the model behaviors in the tension and compression branches separately, which can be desirable in constitutive modeling. In the following, we present representative examples of the generalized strains listed in chronological order.

¹ Since E is a univariate function, we use E' and E'' to denote its first and second derivatives.

Seth–Hill strain family. The Seth–Hill, or Doyle–Ericksen, strain family [63,64] is defined as

$$E(\lambda) = E_{\text{SH}}^{(m)}(\lambda) := \begin{cases} \frac{1}{m}(\lambda^m - 1), & m \neq 0, \\ \ln \lambda, & m = 0. \end{cases}$$

It is the first instantiation of Hill's generalized strain definition and is parameterized by a single parameter. This strain family incorporates several classical strains, including the Almansi ($m = -2$), Hill ($m = -1$), Hencky or logarithmic ($m = 0$), Biot ($m = 1$), and Green–Lagrange ($m = 2$) strains. Except for the Hencky strain, the Seth–Hill strain family does not satisfy the coercive property (2.6) (see Fig. 1), which is viewed as its major deficiency and inspires the subsequent development of new generalized strains.

Curnier–Rakotomanana strain family. In the early 1990s, Curnier and Rakotomanana [70] further generalized the Seth–Hill strain family as

$$E(\lambda) = E_{\text{CR}}^{(m,n)}(\lambda) := \frac{1}{m+n}(\lambda^m - \lambda^{-n}) = \frac{m}{m+n}E_{\text{SH}}^{(m)}(\lambda) + \frac{n}{m+n}E_{\text{SH}}^{(-n)}(\lambda).$$

The authors considered m and n as integer parameters, taking the value of 1 or 2 [70]. This essentially integrates the Pelzer ($m = n = 1$), Mooney ($m = n = 2$), Wall ($m = 1, n = 2$), and Rivlin ($m = 2, n = 1$) strains into a unified format. Since the newly incorporated strains are named after rubber elasticians, it is also termed as the ‘rubber’ family. Later, Darijani and Naghdabadi extended the definition by allowing the two parameters to take non-negative real values with $mn > 0$ [82]. The Hencky strain is supplemented as a natural consequence of L'Hôpital's rule when $m = n = 0$. It can be shown that the scale function satisfies the coercive property (2.6). Also, the two parameters allow one to separately describe the tension and compression material behavior. Notably, studies show that this generalized strain family leads to satisfactory quality of fit for several soft materials [71].

Bažant–Itskov strain family. In the late 1990s, Bažant introduced a family of generalized strains which can now be represented as

$$E(\lambda) = E_{\text{BI}}^{(m)}(\lambda) := \frac{1}{2m}(\lambda^m - \lambda^{-m}) \quad \text{with } m \neq 0.$$

His original goal was to improve the asymptotic approximation of the Hencky strain over the Seth–Hill strains [72]. With a different motivation, Itskov derived the same form based on an analysis of an elastoplasticity problem [73]. Again, the Hencky strain can be supplemented to complete the definition when $m = 0$. This family possesses the coercive and symmetry properties. Apparently, it is a sub-family of the previous Curnier–Rakotomanana strain family when the two parameters are identical. It is also worth mentioning that $\sum_{a=1}^3 E_{\text{BI}}^{(1)}(\lambda_a) \mathbf{n}_a \otimes \mathbf{n}_a$ leads to the two-point tensor introduced by Böck and Holzapfel [83]. Despite several appealing mathematical attributes, the lack of objectivity limits the applicability of the two-point tensor [83] in constitutive modeling [84].

Curnier–Zysset strain family. In 2006, Curnier and Zysset proposed a strain family that is easy to compute and approximate the Seth–Hill family [74]. It can be fitted into the format of the generalized strain (2.3) as

$$E(\lambda) = E_{\text{CZ}}^{(m)}(\lambda) := \frac{2+m}{8}\lambda^2 - \frac{2-m}{8}\lambda^{-2} - \frac{m}{4}, \quad \text{with } -2 \leq m \leq 2.$$

In their study, the well-posedness of the hyperelastic model based on this strain family was analyzed by delimiting the stretch region where the rank-one convexity holds. It was found that this generalized strain family may substantially extend the region for the model to remain rank-one convex, in comparison with the Green–Lagrange strain. This manifests the theoretical significance of developing new concepts of strains.

Darijani–Naghdabadi exponential strain family. In 2013, Darijani and Naghdabadi [75] introduced a strain family as

$$E(\lambda) = E_{\text{DN}}^{(m,n)}(\lambda) := \frac{1}{m+n} \left(e^{m(\lambda-1)} - e^{n(\lambda^{-1}-1)} \right), \quad \text{with } m, n > 0.$$

It can be viewed as an extension of the two-parameter Curnier–Rakotomanana strain family using an exponential function form. This choice can be propitious for characterizing the non-Gaussian region of rubber-like materials [59] and the strain hardening in biological tissues [85].

In our following discussion, we use E as a generic strain without specifying its type. When it becomes necessary, a subscript and superscript are used to indicate the specific strain. For example, $E_{\text{SH}}^{(2)}$ designates the strain of the Seth–Hill family with the parameter $m = 2$ (a.k.a. the Green–Lagrange strain). In Fig. 1, five representative scale functions are depicted, with the symmetry and coercive properties indicated by markers.

Remark 1. The definition (2.3) given here is based on the scale function E of the principal stretch. The generalized strain can be equivalently represented in terms of the stretch tensor $\mathbf{U} := \sqrt{\mathbf{C}}$, which may be convenient in computation. For example, the Curnier–Zysset strain family was originally defined as

$$E_{\text{CZ}}^{(m)} := \frac{2+m}{8}\mathbf{C} - \frac{2-m}{8}\mathbf{C}^{-1} - \frac{m}{4}\mathbf{I},$$

and their motivation for the design partly comes from avoiding spectral decompositions in practical calculations.

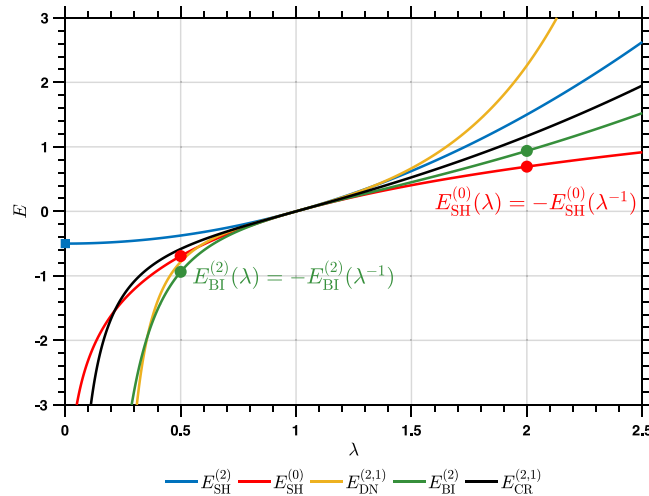


Fig. 1. Illustration of different scale functions. The Green–Lagrange strain (blue curve) gives a finite value when the stretch approaches zero. The Hencky scale function (red curve) and the Bažant–Itskov scale function (green curve) satisfy the symmetry property (2.7), as is indicated by the red and green dots, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.2. Projection tensors

In this section, we introduce three projection tensors that are necessary for constitutive modeling. Considering three distinct principal stretches $\{\lambda_a\}$ for $a = 1, 2, 3$, the derivative of E with respect to $\mathbf{C}$ leads to a rank-four tensor as

$$\mathbb{Q} := 2 \frac{\partial E}{\partial \mathbf{C}} = \sum_{a=1}^3 d_a \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \vartheta_{ab} \mathbf{M}_a \odot \mathbf{M}_b, \quad (2.8)$$

in which

$$d_a := 2 \frac{\partial E}{\partial (\lambda^2)}(\lambda_a) = \frac{1}{\lambda_a} E'(\lambda_a), \quad \vartheta_{ab} := 2 \frac{E(\lambda_a) - E(\lambda_b)}{\lambda_a^2 - \lambda_b^2}, \quad (2.9)$$

and the symbol $\odot$ represents the following tensorial operation (see, e.g., [65, p. 254])

$$\mathbf{M}_a \odot \mathbf{M}_b := \frac{1}{2} (\mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_b + \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_b \otimes \mathbf{N}_a). \quad (2.10)$$

When two or three principal stretches are equal, the difference quotient in (2.9)₂ is replaced by

$$\lim_{\lambda_b \rightarrow \lambda_a} \vartheta_{ab} = d_a,$$

with the aid of L'Hôpital's rule. Here we state and prove a useful result related to $\mathbb{Q}$.

Lemma 1. *There exists a rank-four tensor $\mathbb{Q}^{-1}$ such that $\mathbb{Q} : \mathbb{Q}^{-1} = \mathbb{Q}^{-1} : \mathbb{Q} = \mathbb{I}$.*

Proof. The monotonicity and smoothness of the scale function E guarantee that its inverse E^{-1} exists and is differentiable, due to the inverse function theorem. This indicates that the derivative $\partial \mathbf{C} / \partial E$ is well-defined. We may explicitly construct $\mathbb{Q}^{-1}$ as

$$\mathbb{Q}^{-1} = \sum_{a=1}^3 \frac{\lambda_a}{E'(\lambda_a)} \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \frac{\lambda_a^2 - \lambda_b^2}{2(E(\lambda_a) - E(\lambda_b))} \mathbf{M}_a \odot \mathbf{M}_b.$$

When there are stretches taking identical values, the difference quotient term in the above formula needs to be replaced by $1/d_a$. One may conveniently verify the following identities,

$$\begin{aligned} (\mathbf{M}_a \otimes \mathbf{M}_a) : (\mathbf{M}_b \otimes \mathbf{M}_b) &= \begin{cases} \mathbb{Q}, & a \neq b, \\ \mathbf{M}_a \otimes \mathbf{M}_a, & a = b, \end{cases} \\ (\mathbf{M}_a \odot \mathbf{M}_b) : (\mathbf{M}_c \odot \mathbf{M}_d) &= \begin{cases} \frac{1}{2} \mathbf{M}_a \odot \mathbf{M}_b, & a = c \neq b = d, \\ \frac{1}{2} \mathbf{M}_a \odot \mathbf{M}_b, & a = d \neq b = c, \\ \mathbb{Q}, & \text{otherwise with } a \neq b, c \neq d, \end{cases} \\ (\mathbf{M}_a \otimes \mathbf{M}_a) : (\mathbf{M}_b \odot \mathbf{M}_c) &= \mathbb{Q}, \quad b \neq c. \end{aligned}$$

With the above results, direct calculation of the contraction gives

$$\mathbb{Q} : \mathbb{Q}^{-1} = \mathbb{Q}^{-1} : \mathbb{Q} = \sum_{a=1}^3 \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \mathbf{M}_a \odot \mathbf{M}_b = \mathbb{I},$$

which completes the proof. $\square$

The second derivative of E with respect to C yields a rank-six tensor

$$\begin{aligned} \mathbb{L} &:= 2 \frac{\partial \mathbb{Q}}{\partial C} = 4 \frac{\partial^2 E}{\partial C \partial C} \\ &= \sum_{a=1}^3 f_a \mathbf{M}_a \otimes \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \xi_{ab} (\mathbb{H}_{abb} + \mathbb{H}_{bab} + \mathbb{H}_{bba}) + \sum_{a=1}^3 \sum_{b \neq a}^3 \sum_{c \neq b}^3 \eta \mathbb{H}_{abc}, \end{aligned} \quad (2.11)$$

with

$$f_a := 4 \frac{\partial^2 E}{\partial(\lambda^2) \partial(\lambda^2)}(\lambda_a) = -\frac{1}{\lambda_a^3} E'(\lambda_a) + \frac{1}{\lambda_a^2} E''(\lambda_a), \quad (2.12)$$

$$\xi_{ab} := \frac{\vartheta_{ab} - d_b}{2(\lambda_a^2 - \lambda_b^2)}, \quad (2.13)$$

$$\eta := \sum_{a=1}^3 \sum_{b \neq a}^3 \sum_{c \neq b}^3 \frac{E(\lambda_a)}{2(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)}, \quad (2.14)$$

$$\begin{aligned} \mathbb{H}_{abc} &:= \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_c \otimes \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_c + \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_c \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_b \\ &\quad + \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_b \otimes \mathbf{N}_c + \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_c \otimes \mathbf{N}_b \\ &\quad + \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_a \otimes \mathbf{N}_b \otimes \mathbf{N}_c + \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_b \\ &\quad + \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_b \otimes \mathbf{N}_c + \mathbf{N}_b \otimes \mathbf{N}_a \otimes \mathbf{N}_a \otimes \mathbf{N}_c \otimes \mathbf{N}_c \otimes \mathbf{N}_b. \end{aligned} \quad (2.15)$$

When there exist identical stretches, the coefficients in (2.13) and (2.14) are replaced by the following

$$\lim_{\lambda_b \rightarrow \lambda_a} \xi_{ab} = \frac{1}{8} f_a, \quad \lim_{\substack{\lambda_b \rightarrow \lambda_a \\ \lambda_c \neq \lambda_a}} \eta = \xi_{ca}, \quad \text{and} \quad \lim_{\substack{\lambda_b \rightarrow \lambda_a \\ \lambda_c \rightarrow \lambda_a}} \eta = \frac{1}{8} f_a.$$

The formulas of $\mathbb{Q}$ and $\mathbb{L}$ can be derived based on the principal axes method [86]. In that method, a rate formulation is established and the components of the projection tensors are identified by equating coefficients [65, pp. 258–259]. In this work, we use an alternative approach and derive the formulas by straightforwardly considering the derivatives of the principal axes with respect to the deformation tensor. Our derivations are detailed in Appendices A and B, respectively. The first and second derivatives of $\tilde{E}$ with respect to $\tilde{C}$ are denoted as

$$\tilde{\mathbb{Q}} := 2 \frac{\partial \tilde{E}}{\partial \tilde{C}} \quad \text{and} \quad \tilde{\mathbb{L}} := 4 \frac{\partial^2 \tilde{E}}{\partial \tilde{C} \partial \tilde{C}} = 2 \frac{\partial \tilde{\mathbb{Q}}}{\partial \tilde{C}}. \quad (2.16)$$

The projections $\tilde{\mathbb{Q}}$ and $\tilde{\mathbb{L}}$ can be explicitly represented by the formulas (2.8) and (2.11) with the coefficients evaluated at $\tilde{\lambda}_a$.

The partial derivative of $\tilde{C}$ with respect to C reads

$$\frac{\partial \tilde{C}}{\partial C} := J^{-\frac{2}{3}} \mathbb{P}^T, \quad \text{with} \quad \mathbb{P} := \mathbb{I} - \frac{1}{3} C^{-1} \otimes C. \quad (2.17)$$

The projection tensor $\mathbb{P}$ furnishes the deviatoric behavior in the Lagrangian description [65, p. 230].

2.3. Hyperelasticity of Hill's class

The design of the strain energy function is the crux of material modeling. Different from phenomenological models based on invariants or principal stretches, the hyperelastic model of Hill's class maintains the quadratic functional form while describing nonlinear response by using a generalized strain. Based on this philosophy, the hyperelastic strain energy $\Psi(C)$ can be represented as

$$\Psi(C) = \frac{1}{2} E : \mathbb{E} : E, \quad (2.18)$$

where E is a generalized Lagrangian strain, and $\mathbb{E}$ is a rank-four stiffness tensor. The second Piola–Kirchhoff stress is given by

$$S = 2 \frac{\partial \Psi}{\partial C} = T : \mathbb{Q}, \quad \text{with} \quad T = \frac{\partial \Psi}{\partial E} = \mathbb{E} : E.$$

If the material is isotropic, the stiffness tensor $\mathbb{E}$ takes the form

$$\mathbb{E} = 2\mu \left(\mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \right) + \kappa \mathbf{I} \otimes \mathbf{I}.$$

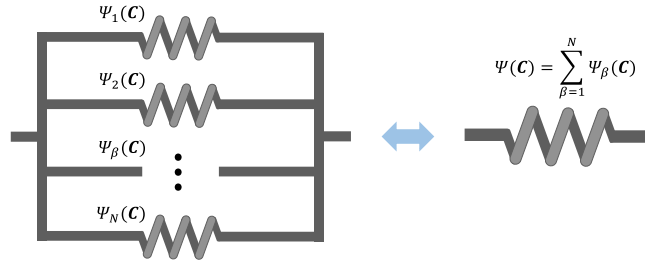


Fig. 2. The interpretation of the generalized hyperelastic strain energy function (2.22) as a device of multiple springs of Hookean type in parallel. Each one of the Hookean spring is described by a single quadratic energy of a specific strain. Their combined mechanical behavior can be viewed as a single spring whose energy is the sum of quadratic strain energies of different strains.

Consequently, one has

$$\Psi(C) = \mu |\text{dev} E|^2 + \frac{\kappa}{2} (\text{tr} E)^2 \quad \text{and} \quad T = 2\mu \text{dev} E + \kappa (\text{tr} E) I, \quad (2.19)$$

where $\text{tr}(\cdot) := (\cdot) : I$ and $\text{dev}(\cdot) := (\cdot) - \text{tr}(\cdot) I/3$. Due to the quadratic form of the energy, the conjugate pair of T and E maintains the Hookean stress–strain relationship. The second Piola–Kirchhoff stress is obtained from the stress-like tensor T by invoking the projection tensor $\mathbb{Q}$ previously defined in (2.8), and this procedure is sometimes referred to as the geometric postprocessing [51]. With a specific choice of the strain E , the above furnishes a specific hyperelasticity model of Hill's class.

It has also been recommended to modify the strain energy (2.19)₁ by replacing $\text{tr} E$ with $\ln(J)$ [87], leading to the following potential,

$$\Psi(C) = \mu |\tilde{E}|^2 + \Psi_{\text{vol}}(J), \quad (2.20)$$

with $\Psi_{\text{vol}}(J) = \kappa \ln(J)^2/2$. This replacement can also be interpreted by instantiating E as the Hencky strain in $\text{tr} E$ [88]. This modified model has been recognized to enjoy improved material stability [65, p. 251]. If the strain in (2.20) is specialized to one of the Seth–Hill strain family, the first term in (2.20) can be expressed as

$$\mu |\tilde{E}_{\text{SH}}^{(m)}|^2 = \frac{\mu}{m^2} \sum_{a=1}^3 (\tilde{\lambda}_a^m - 1)^2, \quad (2.21)$$

which closely resembles the isochoric elastic response function of the Ogden model [65, p. 237].

We notice that there is only one term, or two material parameters, involved in defining the potential (2.21). Considering the Ogden model used for practical data fitting, three terms, or six parameters, are typically needed to provide high quality of fit to experimental data [59,89]. Thereby, there is a need to enrich the form of the aforementioned Hill's hyperelasticity model by introducing additional quadratic strain energies of different strains into the free energy. This strategy has been explored by using multiple Curnier–Rakotomanana strains [71] or Seth–Hill strains [78], both of which demonstrated improved performance in the nonlinear data fitting. Here, we consider an energy represented as

$$\Psi(C) = \sum_{\beta=1}^N \Psi_{\beta}(C), \quad \text{with} \quad \Psi_{\beta}(C) := \frac{1}{2} E_{\beta} : \mathbb{E}_{\beta} : E_{\beta}. \quad (2.22)$$

In the above, the strain energy Ψ is constructed as the sum of multiple quadratic energies whose associated strains E_{β} can be any member of any generalized strain family outlined in Section 2.1. The number of quadratic energy terms is N . An interpretation of this generalization is made in Fig. 2. As a generalization of the modified strain energy (2.20), we consider the potential as

$$\Psi(C) = \sum_{\beta=1}^N \mu_{\beta} |\tilde{E}_{\beta}|^2 + \Psi_{\text{vol}}(J). \quad (2.23)$$

With (2.21), we may view the above strain energy formally similar to the N -term Ogden model in the sense that N different nonlinear functions of stretch constitute the overall potential. Henceforth, we will refer to the model given by (2.18) or (2.20) as the conventional Hill's hyperelasticity, and the elastic model given by (2.22) or (2.23) will be referred to as the generalized Hill's hyperelasticity with multiple terms.

Remark 2. Within the framework of Hill's hyperelasticity, the most well-known example is the Saint Venant–Kirchhoff model, which is instantiated by choosing the Green–Lagrange strain. As illustrated in Part I, it also leads to the finite linear viscoelasticity model [3,10]. However, it is known to fail the polyconvex condition [12], raising concerns about its theoretical validity. Another example is the quadratic Hencky model, widely used in elasticity and inelasticity. Although it is not polyconvex everywhere, it satisfies the Baker–Ericksen inequalities and Hill's inequality [90]. As was summarized in Section 2.1, recent years have witnessed the proposal of more strains, opening the door to the design of models with improved mathematical validity.

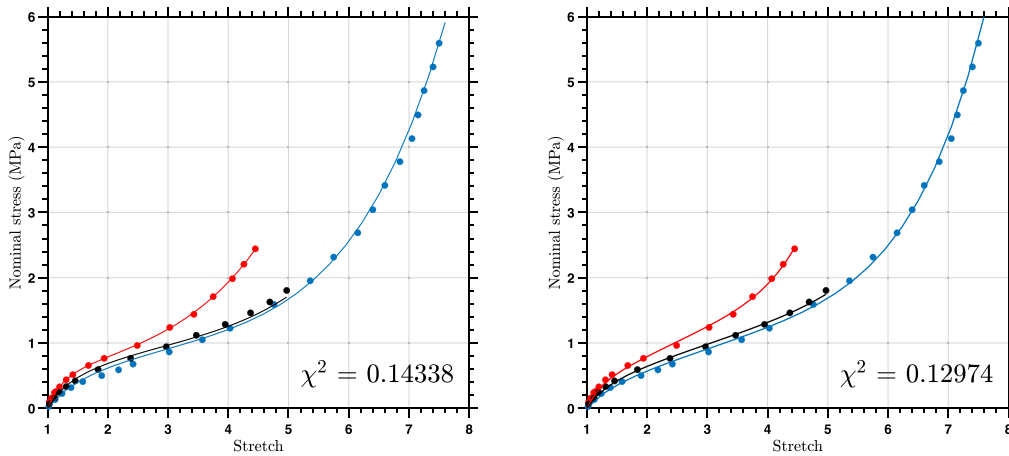


Fig. 3. Fitting of the Ogden model (left) and the model of Hill's class using two Curnier–Rakotomanana strains (right). The dots and lines represent the experimental data and model predictions, respectively. The blue, red, and black colors represent the uniaxial tensile, equi-biaxial tensile, and pure shear responses, respectively. The quality of fit metric χ^2 is adopted from the work of Dal [59, Eqn. (179)]. A smaller value of χ^2 indicates a better model fit. (For interpretation of the references to color, the reader is referred to the web version of this article.)

Remark 3. The efficacy of the proposed model is assessed through the simultaneous fitting of material parameters from a combined suite of uniaxial tensile, equi-biaxial tensile, and pure shear tests of vulcanized rubber, and the experimental data are adopted from the seminal work of Treloar [91]. The model we considered is based on the strain energy of the form (2.22) with $N = 2$, and the two strains are both taken from the Curnier–Rakotomanana family. The three-term Ogden model is considered for comparison purposes. Both models involve six parameters. The fitting results are illustrated in Fig. 3 and demonstrate that the newly proposed model exhibits superior quality of fit to the Ogden model. This indicates that the generalization of Hill's hyperelasticity using multiple terms indeed possesses the potential to improve the capability in describing elastic behavior under large deformations.

2.4. Kinematic assumptions of the Green–Naghdi type

In the original work of Green and Naghdi [6,7,44], a fundamental assumption is the existence of a symmetric Lagrangian tensor known as the plastic strain, and the difference between the total strain and the plastic strain enters into the stress response function. This strategy was later generalized with the invocation of Seth–Hill strains. Inspired by this approach and new strain families developed recently, we construct a nonlinear viscoelasticity theory based on similar assumptions. Our *first* assumption is the existence of a rank-two Lagrangian tensor $\Gamma \in \text{Sym}(3)_+$. It is a primitive variable characterizing the viscous deformation and is analogous to the deformation tensor C . As a symmetric tensor, the spectral decomposition of Γ can be performed, leading to

$$\Gamma := \sum_{a=1}^3 \Gamma_a^2 \tilde{M}_a, \quad \tilde{M}_a := \tilde{N}_a \otimes \tilde{N}_a, \quad (2.24)$$

where $\{\Gamma_a^2\}$ and $\{\tilde{N}_a\}$, for $a = 1, 2, 3$, are the eigenvalues and eigenvectors of Γ , respectively. We mention that Γ is not necessarily coaxial with C , and the “bar” notation in $\{\tilde{N}_a\}$ is used to distinguish the axes from $\{N_a\}$. With the decomposition (2.24), we may introduce the corresponding “viscous strain” $E^v \in \text{Sym}(3)_+$ as

$$E^v := \sum_{a=1}^3 E^v(\Gamma_a) \tilde{M}_a. \quad (2.25)$$

We mention that E^v in (2.25) is a generic scale function and does not necessarily coincide with the scale function used in the total strain E . The *second* kinematic assumption is that $E - E^v$ enters into the constitutive formulation. We further introduce the following tensors

$$\mathbb{Q}^v := 2 \frac{\partial E^v}{\partial \Gamma} \quad \text{and} \quad \mathbb{L}^v := 4 \frac{\partial^2 E^v}{\partial \Gamma \partial \Gamma} = 2 \frac{\partial \mathbb{Q}^v}{\partial \Gamma}, \quad (2.26)$$

which are the viscous analogues of $\mathbb{Q}$ and $\mathbb{L}$. The explicit representations of $\mathbb{Q}^v$ and $\mathbb{L}^v$ follow the formula (2.8) and (2.11) with the coefficients evaluated at Γ_a . We immediately have the following result, whose proof follows that of Lemma 1.

Lemma 2. *There exists a rank-four tensor $\mathbb{Q}^{v-1}$ such that $\mathbb{Q}^v : \mathbb{Q}^{v-1} = \mathbb{Q}^{v-1} : \mathbb{Q}^v = \mathbb{I}$.*

The internal state variable Γ is introduced to describe the non-equilibrium behavior of the material. Conceptually, it can be viewed as a viscous deformation-like tensor associated with the dashpot when using a spring–dashpot rheological model for interpretation (see Fig. 4). For typical viscoelastic solids, their non-equilibrium behavior may involve multiple relaxation processes,

meaning there can be multiple Maxwell elements arranged in parallel in the rheological model. It thus requires multiple independent internal state variables to adequately model the material behavior. In our discussion, we construct the constitutive theory based on a single relaxation process using Γ , and the extension to M relaxation processes can be made with a set of internal state variables $\{\Gamma^\alpha\}_{\alpha=1}^M$. Without considering thermal effects, the *equilibrium state* can be characterized kinematically by imposing that the rates of change of the internal state variables are identically zero. Here, we provide a rigorous definition for this notion.

Definition 1. For a viscoelastic material characterized by M internal state variables $\{\Gamma^\alpha\}_{\alpha=1}^M$, it is in the equilibrium state if

$$\frac{d}{dt}\Gamma^\alpha = \mathbf{0}, \quad (2.27)$$

for $\alpha = 1, \dots, M$.

We adopt the notation $|_{\text{eq}}$ introduced in Part I to indicate quantities evaluated at the equilibrium state. In an alternative way to characterize the equilibrium state, the force driving the viscous deformation is demanded to vanish in the equilibrium limit [11, p. 345]. We mention that the two conditions are not necessarily equivalent in the *finite strain* setting, as we did encounter non-vanishing non-equilibrium stresses in certain models that caused pathological behavior previously (see Section 4.1 in Part I). Nevertheless, we will show in the subsequent discussion that, with the kinematic assumption of the Green–Naghdi type, the non-equilibrium stress can be rigorously proved to be fully relaxed in the equilibrium state.

Remark 4. It is well-known that the difference $E - E^v$ is not necessarily an elastic strain. Here, we use E^e simply as a notation for this difference without providing additional interpretations. The normality condition (2.5)₂ of the generalized strain guarantees that it can be approximated by an infinitesimal strain in the small strain limit. The second kinematic assumption, together with the generalized strain concept, naturally recovers the decomposition used in the infinitesimal strain inelasticity theory, that is, $\epsilon = \epsilon^v + \epsilon^e$ [11], in which ϵ , ϵ^e and ϵ^v are the infinitesimal approximations of E , E^v , and E^e , respectively.

Remark 5. When it becomes necessary, we use $E^v(\Gamma^\alpha)$ to clarify the dependency of a generalized strain on the deformation-like tensor (the α th internal state variable Γ^α here). In practice, we leverage a robust and accurate algorithm [92–94] to perform spectral decompositions for C and $\{\Gamma^\alpha\}_{\alpha=1}^M$, which are then utilized to construct the generalized strains based on the corresponding scale functions.

Remark 6. In a different approach, the kinematic assumption postulates the multiplicative decomposition $F = F^e F^v$ based on the concept of the intermediate configuration [13,15,16]. One may establish the equivalence between the proposed theory and the multiplicative theory using specific strain or under specific deformation states. For example, if the internal state variable Γ is interpreted as $C^v := F^{vT} F^v$, there is an equivalence of the elastic strain invariants using the Green–Lagrange strain. Further elaborations on these insights can be found in Appendix C, shedding lights on the connections between the two fundamentally distinct kinematic assumptions.

2.5. Constitutive theory based on the Helmholtz free energy

In this section, we derive constitutive relations based on the Helmholtz free energy to provide a theory that works within the classical pure displacement formulation. Our discussion starts by considering a single relaxation process and using conventional Hill’s hyperelasticity. Under the isothermal condition, the Helmholtz free energy Ψ is postulated as a function of C and Γ , that is, $\Psi = \Psi(C, \Gamma)$. Due to the definitions of generalized strains, the potential Ψ can be equivalently expressed as a function of $E = E(C)$ and $E^v = E^v(\Gamma)$. A number of experiments suggest the existence of distinct hyperelastic and time-dependent contributions of the energy [95,96], leading to the following form of the free energy

$$\Psi(C, \Gamma) = \Psi^\infty(E) + Y(E - E^v). \quad (2.28)$$

The superscript ‘ ∞ ’ represents the contribution arising from the material’s equilibrium response. The potential Y is associated with the dissipative behavior and is known as the configurational free energy [1,65]. Sometimes, it is also referred to as the non-equilibrium part of the free energy [16], and we use the superscript ‘ neq ’ to indicate quantities related to Y . We mention that hyperelasticity is modeled following Hill’s framework, and the energies Ψ^∞ and Y are, respectively, quadratic functions of E and $E - E^v$. They may take the functional form of either (2.18) or (2.20), and we do not specify their explicit forms at this stage. After function composition, we may express the two potentials as $\Psi^\infty(C)$ and $Y(C, \Gamma)$ without distinguishing the functional form. The significance of (2.28) is that the two deformation tensors C and Γ enter into the potential Y through $E(C) - E^v(\Gamma)$, exemplifying the second kinematic assumption of the Green–Naghdi type. The Clausius–Planck inequality reads

$$D := S : \frac{1}{2} \frac{dC}{dt} - \frac{d\Psi}{dt} \geq 0, \quad (2.29)$$

where D represents the internal dissipation. With the postulated potential form (2.28), the inequality can be expanded as

$$D = \left(S - 2 \frac{\partial \Psi}{\partial C} \right) : \frac{1}{2} \frac{dC}{dt} + Q : \frac{1}{2} \frac{d\Gamma}{dt} \geq 0,$$

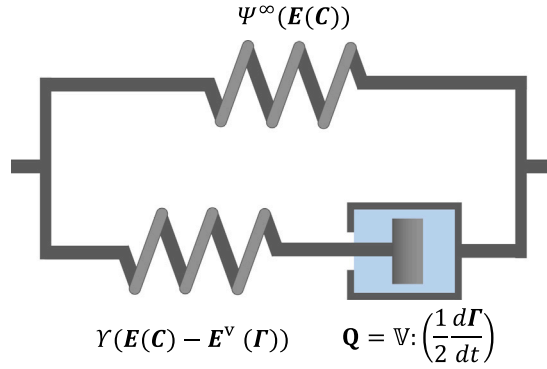


Fig. 4. An illustration of the theory based on the Helmholtz free energy using the spring–dashpot rheological model.

with

$$\mathbf{Q} := -2 \frac{\partial \Psi}{\partial \Gamma}. \quad (2.30)$$

To ensure the satisfaction of the inequality for arbitrary kinematic processes, we make the following choices,

$$\mathbf{S} := 2 \frac{\partial \Psi}{\partial \mathbf{C}}, \quad \mathbf{Q} = \mathbb{V} : \left(\frac{1}{2} \frac{d\Gamma}{dt} \right), \quad (2.31)$$

with $\mathbb{V}$ being a positive-definite rank-four viscosity tensor. It is additionally assumed that there exists $\mathbb{V}^{-1}$ such that the constitutive relation (2.31)₂ can be equivalently expressed as

$$\frac{1}{2} \frac{d\Gamma}{dt} = \mathbb{V}^{-1} : \mathbf{Q}. \quad (2.32)$$

With the above constitutive relations, the internal dissipation becomes

$$D = \left(\frac{1}{2} \frac{d\Gamma}{dt} \right) : \mathbb{V} : \left(\frac{1}{2} \frac{d\Gamma}{dt} \right) = \mathbf{Q} : \mathbb{V}^{-1} : \mathbf{Q}, \quad (2.33)$$

which remains non-negative due to the positive-definiteness of the viscosity tensor $\mathbb{V}$. An illustration of this nonlinear viscoelasticity model is made through the spring–dashpot device in Fig. 4.

Leveraging the split of the Helmholtz free energy (2.28), the second Piola–Kirchhoff stress $\mathbf{S}$ can be decomposed into equilibrium and non-equilibrium parts,

$$\mathbf{S} = \mathbf{S}^\infty + \mathbf{S}^{\text{neq}}, \quad (2.34)$$

with

$$\mathbf{S}^\infty := \mathbf{T}^\infty : \mathbb{Q}, \quad \mathbf{S}^{\text{neq}} := \mathbf{T}^{\text{neq}} : \mathbb{Q}, \quad \mathbf{T}^\infty := \frac{\partial \Psi^\infty}{\partial \mathbf{E}}, \quad \text{and} \quad \mathbf{T}^{\text{neq}} := \frac{\partial Y}{\partial \mathbf{E}} = -\frac{\partial Y}{\partial E^v}. \quad (2.35)$$

In the above, $\mathbf{T}^\infty$ and $\mathbf{T}^{\text{neq}}$ are the work conjugate stresses to the generalized strain $\mathbf{E}$ with respect to the equilibrium and non-equilibrium potentials. We also recognize that the definition (2.30) of $\mathbf{Q}$ can be refined as

$$\mathbf{Q} = -2 \frac{\partial Y}{\partial \Gamma} = -\frac{\partial Y}{\partial E^v} : 2 \frac{\partial E^v}{\partial \Gamma} = \mathbf{T}^{\text{neq}} : \mathbb{Q}^v. \quad (2.36)$$

With the constitutive relation (2.31)₂, it can be readily shown that $\mathbf{Q}|_{\text{eq}} = \mathbf{O}$. A well-posed model needs to ensure $\mathbf{S}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$ in the equilibrium state (see also Remark 1 of [16]). An intuitive interpretation is that the force driving the dashpot vanishes in the equilibrium state (refer to Fig. 4). In the above theory, $\mathbf{S}^{\text{neq}} \neq \mathbf{Q}$ simply because the two projection tensors $\mathbb{Q}$ and $\mathbb{Q}^v$ are not necessarily identical. Nevertheless, we still can show the vanishment of $\mathbf{S}^{\text{neq}}$ in the following proposition.

Proposition 1. *In the equilibrium state, the non-equilibrium stress vanishes, i.e.,*

$$\mathbf{S}^{\text{neq}}|_{\text{eq}} = \mathbf{O}.$$

Proof. According to Lemma 1, there exists a rank-four tensor $\mathbb{Q}^{v-1}$ such that $\mathbb{Q}^v : \mathbb{Q}^{v-1} = \mathbb{I}$. Therefore, one has the following relation

$$\mathbf{Q} : \mathbb{Q}^{v-1} = \mathbf{T}^{\text{neq}} : \mathbb{Q}^v : \mathbb{Q}^{v-1} = \mathbf{T}^{\text{neq}}.$$

It can be shown that $\mathbf{Q}|_{\text{eq}} = \mathbf{O}$ due to (2.31)₂ and the definition of the equilibrium state given in (2.27). Consequently one has $\mathbf{T}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$, which further leads to $\mathbf{S}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$. $\square$

Remark 7. If the viscosity tensor $\mathbb{V}$ is isotropic, it can be represented in the following form,

$$\mathbb{V} = 2\eta_d \left(\mathbb{I} - \frac{1}{3} \mathbf{I} \otimes \mathbf{I} \right) + \frac{2}{3} \eta_v \mathbf{I} \otimes \mathbf{I},$$

with η_d and η_v being the deviatoric and volumetric viscosities, respectively. If the two coefficients are identical (i.e., $\eta = \eta_d = \eta_v$), the constitutive relation (2.31)₂ and the internal dissipation get further simplified as

$$\frac{d\mathbf{F}}{dt} = \frac{1}{\eta} \mathbf{Q} \quad \text{and} \quad D = \frac{\eta}{2} \left| \frac{d\mathbf{F}}{dt} \right|^2 = \frac{1}{2\eta} |\mathbf{Q}|^2.$$

2.5.1. Generalization: Hill's hyperelasticity of multiple terms and multiple relaxation processes

As was discussed in Section 2.3, the capability of conventional Hill's hyperelasticity in characterizing nonlinear material behavior can be improved by adopting multiple quadratic terms. Moreover, we assume that there are M independent internal state variables $\{\mathbf{F}^\alpha\}_{\alpha=1}^M$, and each one of them is associated with a positive-definite viscosity tensor $\mathbb{V}^\alpha$. Based on this generalization strategy, the two potentials Ψ^∞ and Y can be conveniently modified into the sums of multiple quadratic terms,

$$\Psi^\infty(\mathbf{C}) = \sum_{\beta=1}^N \Psi_\beta^\infty(\mathbf{C}), \quad Y(\mathbf{C}, \mathbf{F}^1, \dots, \mathbf{F}^M) = \sum_{\alpha=1}^M Y^\alpha(\mathbf{C}, \mathbf{F}^\alpha), \quad (2.37)$$

with Ψ_β^∞ and Y^α being the quadratic function of $\mathbf{E}_\beta^\infty(\mathbf{C})$ and $\mathbf{E}^\alpha(\mathbf{C}) - \mathbf{E}^\alpha(\mathbf{F}^\alpha)$, respectively. The stresses are correspondingly expressed as

$$\mathbf{S}^\infty = \sum_{\beta=1}^N \mathbf{T}_\beta^\infty : \mathbb{Q}_\beta^\infty, \quad \mathbf{S}^{\text{neq}} = \sum_{\alpha=1}^M \mathbf{T}^\alpha : \mathbb{Q}^\alpha, \quad \mathbf{Q}^\alpha := -2 \frac{\partial Y}{\partial \mathbf{F}^\alpha} = -2 \frac{\partial Y^\alpha}{\partial \mathbf{F}^\alpha} = \mathbf{T}^\alpha : \mathbb{Q}^{\vee \alpha}, \quad (2.38)$$

with

$$\mathbf{T}_\beta^\infty := \frac{\partial \Psi_\beta^\infty}{\partial \mathbf{E}_\beta^\infty}, \quad \mathbf{T}^\alpha := \frac{\partial Y^\alpha}{\partial \mathbf{E}^\alpha} = - \frac{\partial Y^\alpha}{\partial \mathbf{E}^{\vee \alpha}}, \quad (2.39)$$

$$\mathbb{Q}_\beta^\infty := 2 \frac{\partial \mathbf{E}_\beta^\infty}{\partial \mathbf{C}}, \quad \mathbb{Q}^\alpha := 2 \frac{\partial \mathbf{E}^\alpha}{\partial \mathbf{C}}, \quad \text{and} \quad \mathbb{Q}^{\vee \alpha} := 2 \frac{\partial \mathbf{E}^{\vee \alpha}}{\partial \mathbf{F}^\alpha}. \quad (2.40)$$

For each one of the relaxation processes, the evolution equation reads

$$\mathbf{Q}^\alpha = \mathbb{V}^\alpha : \left(\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} \right) \quad \text{or} \quad \frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} = (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha. \quad (2.41)$$

The internal dissipation becomes

$$D = \sum_{\alpha=1}^M \left(\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} \right) : \mathbb{V}^\alpha : \left(\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} \right) = \sum_{\alpha=1}^M \mathbf{Q}^\alpha : (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha,$$

involving the dissipation from the M different processes. For the generalized case, we show that the property given by Proposition 1 still holds.

Proposition 2. For the generalized model with the potential given by (2.37), the non-equilibrium stress vanishes in the equilibrium state, i.e.,

$$\mathbf{S}^{\text{neq}}|_{\text{eq}} = \mathbf{O}.$$

Proof. Due to the definition of the equilibrium state, we have $\mathbf{Q}^\alpha|_{\text{eq}} = \mathbf{O}$. According to Lemma 1, one has the following relation,

$$\mathbf{Q}^\alpha : \mathbb{Q}^{\vee \alpha -1} = \mathbf{T}^\alpha : \mathbb{Q}^{\vee \alpha} : \mathbb{Q}^{\vee \alpha -1} = \mathbf{T}^\alpha,$$

which implies $\mathbf{T}^\alpha|_{\text{eq}} = \mathbf{O}$. Following the expression for the non-equilibrium stress, it can be shown that

$$\mathbf{S}^{\text{neq}}|_{\text{eq}} = \sum_{\alpha=1}^M \mathbf{T}^\alpha|_{\text{eq}} : \mathbb{Q}^\alpha = \mathbf{O},$$

which completes the proof. $\square$

Remark 8. It remains an interesting topic to further generalize the theory by allowing multiple quadratic terms in the definition of Y^α . However, we found that the result of Proposition 2 will not hold with that generalization.

2.6. Constitutive theory based on the Gibbs free energy

There have been experimental evidences suggesting that the shear viscoelastic effects are more significant than the bulk viscoelasticity for most materials [97, Chapter 18]. Following those observations, we consider the decomposition of the Helmholtz free energy into the volumetric and isochoric parts,

$$\Psi(\mathbf{C}, \mathbf{F}) = \Psi_{\text{vol}}^\infty(J) + \Psi_{\text{iso}}(\tilde{\mathbf{E}}, \mathbf{E}^\vee). \quad (2.42)$$

based on the decomposition of the Flory type (2.1). Following the idea of energy split introduced in (2.28), the isochoric part of the energy Ψ_{iso} gets further decomposed into the equilibrium and non-equilibrium parts,

$$\Psi_{\text{iso}}(\tilde{\mathbf{E}}, \mathbf{E}^v) = \Psi_{\text{iso}}^\infty(\tilde{\mathbf{E}}) + Y(\tilde{\mathbf{E}} - \mathbf{E}^v). \quad (2.43)$$

For the Helmholtz free energy, an issue is the singular behavior as the model approaches the incompressible limit. To circumvent this issue, a Legendre transformation can be performed on the volumetric part of the energy Ψ_{vol}^∞ ,

$$G_{\text{vol}}^\infty(P) := \inf_J \{ \Psi_{\text{vol}}^\infty(J) + PJ \}, \quad (2.44)$$

where P is the thermodynamic pressure defined on the Lagrangian configuration. The resulting Gibbs free energy is defined as

$$G(\tilde{\mathbf{C}}, P, \mathbf{\Gamma}) := G_{\text{vol}}^\infty(P) + G_{\text{iso}}(\tilde{\mathbf{E}}, \mathbf{E}^v) = G_{\text{vol}}^\infty(P) + G_{\text{iso}}^\infty(\tilde{\mathbf{E}}) + Y(\tilde{\mathbf{E}}, \mathbf{E}^v), \quad (2.45)$$

with

$$G_{\text{iso}}^\infty(\tilde{\mathbf{E}}) = \Psi_{\text{iso}}^\infty(\tilde{\mathbf{E}}) = \mu^\infty |\tilde{\mathbf{E}}|^2 \quad \text{and} \quad Y(\tilde{\mathbf{E}} - \mathbf{E}^v) = \mu^{\text{neq}} |\tilde{\mathbf{E}} - \mathbf{E}^v|^2. \quad (2.46)$$

In the above, we replaced the notation Ψ_{iso}^∞ by G_{iso}^∞ to indicate that it constitutes a part of the Gibbs free energy. The Legendre transformation does not affect the isochoric part, meaning G_{iso}^∞ and Y are directly inherited from the Helmholtz free energy. Also, the design of the two potentials here follows the modified quadratic form (2.23), with μ^∞ (μ^{neq}) representing the shear modulus for the (non-)equilibrium part. For notational simplicity, we momentarily consider a single quadratic term for the elastic model here. Due to the Legendre transformation (2.44), the pressure P enters into the theory as a primitive variable, and the subsequent formulation becomes of the saddle-point nature, which is well-behaved in both compressible and incompressible regimes [1,98] (see Fig. 1 in Part I for an interpretation). In fact, one may show that the Gibbs free energy (2.45) leads to a generalization of the Herrmann variational formulation to the finite strain regimes [99–102]. With the Gibbs free energy (2.45), the Clausius–Planck inequality is modified to

$$\begin{aligned} D &= S : \frac{1}{2} \frac{d\mathbf{C}}{dt} - \frac{dG}{dt} + \frac{dP}{dt} J + P \frac{dJ}{dt} \\ &= \left(S - 2 \frac{\partial G_{\text{iso}}^\infty}{\partial \tilde{\mathbf{C}}} : \frac{\partial \tilde{\mathbf{C}}}{\partial \tilde{\mathbf{C}}} - 2 \frac{\partial Y}{\partial \tilde{\mathbf{C}}} : \frac{\partial \tilde{\mathbf{C}}}{\partial \tilde{\mathbf{C}}} + 2P \frac{\partial J}{\partial \tilde{\mathbf{C}}} \right) : \frac{1}{2} \frac{d\mathbf{C}}{dt} + \left(J - \frac{dG_{\text{vol}}^\infty}{dP} \right) \frac{dP}{dt} - 2 \frac{\partial Y}{\partial \mathbf{\Gamma}} : \frac{1}{2} \frac{d\mathbf{\Gamma}}{dt} \\ &= \left(S - J^{-\frac{2}{3}} \mathbb{P} : (\tilde{\mathbf{S}}_{\text{iso}}^\infty + \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}) + PJ\mathbf{C}^{-1} \right) : \frac{1}{2} \frac{d\mathbf{C}}{dt} + \left(J - \frac{dG_{\text{vol}}^\infty}{dP} \right) \frac{dP}{dt} + \mathbf{Q} : \frac{1}{2} \frac{d\mathbf{\Gamma}}{dt}. \end{aligned}$$

In the last equality, we invoked the definition of $\mathbb{P}$ given in (2.17) and introduced three stress-like tensors as

$$\tilde{\mathbf{S}}_{\text{iso}}^\infty := 2 \frac{\partial G_{\text{iso}}^\infty}{\partial \tilde{\mathbf{C}}}, \quad \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}} := 2 \frac{\partial Y}{\partial \tilde{\mathbf{C}}}, \quad \text{and} \quad \mathbf{Q} := -2 \frac{\partial Y}{\partial \mathbf{\Gamma}}. \quad (2.47)$$

Based on the above relation for the internal dissipation D , the following choices are made,

$$\rho = \rho_0 \left(\frac{dG_{\text{vol}}^\infty}{dP} \right)^{-1} \circ \phi_t^{-1}, \quad \mathbf{S} = \mathbf{S}_{\text{vol}} + \mathbf{S}_{\text{iso}} = \mathbf{S}_{\text{vol}} + \mathbf{S}_{\text{iso}}^\infty + \mathbf{S}_{\text{iso}}^{\text{neq}}, \quad \mathbf{Q} = \mathbb{V} : \left(\frac{1}{2} \frac{d\mathbf{\Gamma}}{dt} \right), \quad (2.48)$$

$$\mathbf{S}_{\text{vol}} = -JPC^{-1}, \quad \mathbf{S}_{\text{iso}}^\infty = J^{-\frac{2}{3}} \mathbb{P} : \tilde{\mathbf{S}}_{\text{iso}}^\infty, \quad \mathbf{S}_{\text{iso}}^{\text{neq}} = J^{-\frac{2}{3}} \mathbb{P} : \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}, \quad (2.49)$$

in which $\circ$ denotes function composition and $\mathbb{V}$ is the positive-definite viscosity tensor. Noticing that ρ is typically defined on the current configuration, we invoke ϕ_t^{-1} in (2.48)₁ to make a proper definition of it. With the above constitutive relations, the internal dissipation becomes

$$D = \left(\frac{1}{2} \frac{d\mathbf{\Gamma}}{dt} \right) : \mathbb{V} : \left(\frac{1}{2} \frac{d\mathbf{\Gamma}}{dt} \right) = \mathbf{Q} : \mathbb{V}^{-1} : \mathbf{Q}$$

and remains non-negative for arbitrary kinematic processes. An intuitive interpretation of the isochoric part of this model is made through the spring–dashpot device in Fig. 5. Since the potentials are expressed in terms of $\tilde{\mathbf{E}}$, the stress-like tensors (2.47) are defined in the following refined manner,

$$\tilde{\mathbf{S}}_{\text{iso}}^\infty = \tilde{\mathbf{T}}^\infty : \tilde{\mathbf{Q}}, \quad \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}} = \tilde{\mathbf{T}}^{\text{neq}} : \tilde{\mathbf{Q}}, \quad \mathbf{Q} = \tilde{\mathbf{T}}^{\text{neq}} : \mathbf{Q}^v, \quad (2.50)$$

with

$$\tilde{\mathbf{T}}^\infty := \frac{\partial G_{\text{iso}}^\infty}{\partial \tilde{\mathbf{E}}} = 2\mu^\infty \tilde{\mathbf{E}} \quad \text{and} \quad \tilde{\mathbf{T}}^{\text{neq}} := \frac{\partial Y}{\partial \tilde{\mathbf{E}}} = 2\mu^{\text{neq}}(\tilde{\mathbf{E}} - \mathbf{E}^v). \quad (2.51)$$

Similar to the result of Proposition 1, we prove the relaxation property of the stress associated with the potential Y in the equilibrium state. This property is not obvious because $\mathbf{S}_{\text{iso}}^{\text{neq}}$ given in (2.49)₃ and $\mathbf{Q}$ given in (2.48)₃ are not necessarily identical.

Proposition 3. *In the equilibrium state, the non-equilibrium stress $\mathbf{S}_{\text{iso}}^{\text{neq}}$ vanishes, that is,*

$$\mathbf{S}_{\text{iso}}^{\text{neq}} \Big|_{\text{eq}} = \mathbf{O}.$$

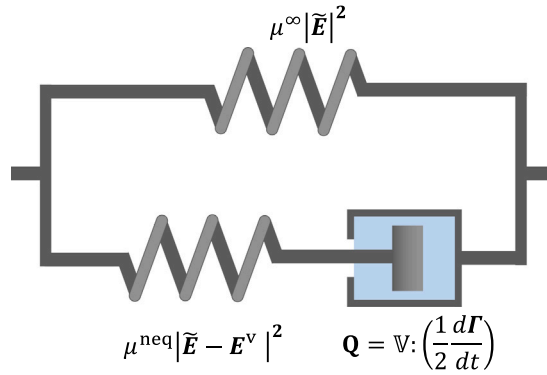


Fig. 5. An illustration of the theory based on the Gibbs free energy using the spring–dashpot rheological model.

Proof. Due to Lemma 2, multiplying $\mathbb{Q}^{\vee-1}$ at both sides of (2.50)₃ leads to

$$\mathbf{Q} : \mathbb{Q}^{\vee-1} = \tilde{\mathbf{T}}^{\text{neq}} : \mathbb{Q}^{\vee} : \mathbb{Q}^{\vee-1} = \tilde{\mathbf{T}}^{\text{neq}}.$$

It is known that $\mathbf{Q}|_{\text{eq}} = \mathbf{O}$. This implies $\tilde{\mathbf{T}}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$, which leads to $\tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}|_{\text{eq}} = \mathbf{S}_{\text{iso}}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$. $\square$

Remark 9. If the strains are taken as the Green–Lagrange strain, we have $\tilde{\mathbb{Q}} = \mathbb{Q}^{\vee} = \mathbb{I}$. Consequently, one has

$$\tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}} = \mathbf{Q} = \mu^{\text{neq}} (\tilde{\mathbf{C}} - \mathbf{\Gamma}),$$

and the configurational free energy Y is identical to the energy (1.2) given at the beginning of this article. If the viscosity tensor $\mathbb{V}$ is isotropic, the derived constitutive model recovers the finite linear viscoelasticity model [3,10] discussed in Part I. In this regard, we may view the model proposed here as the nonlinear generalization of the Holzapfel–Simo–Saint Venant–Kirchhoff model.

Remark 10. In this work, the evolution equation is obtained by combining (2.48)₃ and (2.50)₃, which results in

$$\mathbf{Q} = \mathbb{V} : \left(\frac{1}{2} \frac{d\mathbf{\Gamma}}{dt} \right) = 2\mu^{\text{neq}} (\tilde{\mathbf{E}} - \mathbf{E}^{\vee}) : \mathbb{Q}^{\vee}. \quad (2.52)$$

Noticing that $\mathbf{E}^{\vee}$ generally depends on $\mathbf{\Gamma}$ in a nonlinear manner, the above is a nonlinear evolution equation for $\mathbf{\Gamma}$. In our previous study, we name it as a strain-driven format because the primitive unknown $\mathbf{\Gamma}$ is deformation-like [103]. If both $\tilde{\mathbf{E}}$ and $\mathbf{E}^{\vee}$ are chosen as the Green–Lagrange strain and $\mathbb{V} = 2\eta\mathbb{I}$, the evolution equation simplifies to

$$\mathbf{Q} = \eta \frac{d\mathbf{\Gamma}}{dt} = \mu^{\text{neq}} (\tilde{\mathbf{C}} - \mathbf{\Gamma}).$$

Taking time derivative at both sides yields

$$\frac{d\mathbf{Q}}{dt} + \frac{\mu^{\text{neq}}}{\eta} \mathbf{Q} = \mu^{\text{neq}} \frac{d\tilde{\mathbf{C}}}{dt},$$

which becomes an evolution equation for the stress-like tensor $\mathbf{Q}$. The above stress-driven evolution equation is in fact the most considered type for the finite deformation linear viscoelasticity models [1,4,10,11,104]. Nevertheless, deriving an evolution equation for $\mathbf{Q}$ in the nonlinear model becomes non-trivial. More importantly, working on the evolution equation for the deformation-like tensor $\mathbf{\Gamma}$ may lead to constitutive integration algorithms that preserve critical physical and mathematical structures [103]. In this study, our constitutive integration is based on the strain-driven evolution equation.

Remark 11. When the strains $\mathbf{E}$ and $\mathbf{E}^{\vee}$ are chosen as the Almansi strain and $\mathbb{V} = 2\eta\mathbb{I}$, the evolution Eq. (2.52) can be represented as

$$\frac{d\mathbf{\Gamma}^{-1}}{dt} = \frac{\mu^{\text{neq}}}{\eta} \mathbb{Q}^{\vee} : (\tilde{\mathbf{C}}^{-1} - \mathbf{\Gamma}^{-1}) : \mathbb{Q}^{\vee}.$$

If the projection tensors $\mathbb{Q}^{\vee}$ are ignored in the equation above, the evolution equation originally introduced by Lubliner [105] is recovered. Notably, that evolution equation enjoys sound physical interpretation at the microscopic level [58,106] and is widely used in soft material modeling [107,108]. In fact, it can also be derived from the linearization of the evolution equation of the multiplicative theory [16]. Additionally, in the model of Lubliner, the term $\tilde{\mathbf{C}}\mathbf{\Gamma}^{-1}$ enters into the definition of the configurational free energy [105,108], and its connection with the multiplicative theory is elucidated in Appendix C.

Remark 12. The volumetric energy $\Psi_{\text{vol}}^{\infty}$ is typically convex, guaranteeing the validity of the Legendre transformation. Readers may refer to [98, Sec. 2.4] for a discussion of some commonly used volumetric energies $\Psi_{\text{vol}}^{\infty}$ and their counterparts G_{vol}^{∞} .

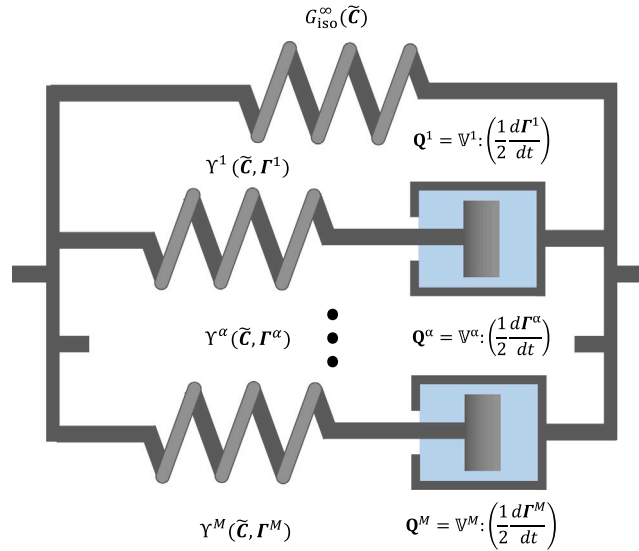


Fig. 6. An illustration of the multiple relaxation processes based on the Gibbs free energy by the spring–dashpot device.

2.6.1. Generalization: Hill's hyperelasticity of multiple terms and multiple relaxation processes

Similar to the generalization made in Section 2.5.1, we consider the hyperelasticity of Hill's class with multiple terms. The free energies G_{iso}^{∞} and Y are modified by using multiple quadratic terms,

$$G_{\text{iso}}^{\infty}(\tilde{\mathbf{C}}) = \sum_{\beta=1}^N G_{\text{iso}\beta}^{\infty}(\tilde{\mathbf{C}}), \quad Y(\tilde{\mathbf{C}}, \Gamma^1, \dots, \Gamma^M) = \sum_{\alpha=1}^M Y^{\alpha}(\tilde{\mathbf{C}}, \Gamma^{\alpha}), \quad (2.53)$$

with

$$G_{\text{iso}\beta}^{\infty}(\tilde{\mathbf{C}}) := \mu_{\beta}^{\infty} \left| \tilde{\mathbf{E}}_{\beta}^{\infty}(\tilde{\mathbf{C}}) \right|^2, \quad Y^{\alpha}(\tilde{\mathbf{C}}, \Gamma^{\alpha}) := \mu^{\alpha} \left| \tilde{\mathbf{E}}^{\alpha}(\tilde{\mathbf{C}}) - \mathbf{E}^{\nu\alpha}(\Gamma^{\alpha}) \right|^2. \quad (2.54)$$

The values of N and M represent the number of quadratic terms used in the equilibrium energy and the number of relaxation processes, respectively. A rheological model is depicted in Fig. 6 to illustrate this generalization. The stresses are modified as

$$\tilde{\mathbf{S}}_{\text{iso}}^{\infty} = \sum_{\beta=1}^N \tilde{\mathbf{T}}_{\beta}^{\infty} : \tilde{\mathbf{Q}}_{\beta}^{\infty}, \quad \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}} = \sum_{\alpha=1}^M \tilde{\mathbf{T}}^{\alpha} : \tilde{\mathbf{Q}}^{\alpha}, \quad \mathbf{Q}^{\alpha} := -2 \frac{\partial Y}{\partial \Gamma^{\alpha}} = -2 \frac{\partial Y^{\alpha}}{\partial \Gamma^{\alpha}} = \tilde{\mathbf{T}}^{\alpha} : \mathbb{Q}^{\nu\alpha}, \quad (2.55)$$

with

$$\tilde{\mathbf{T}}_{\beta}^{\infty} := \frac{\partial G_{\text{iso}\beta}^{\infty}}{\partial \tilde{\mathbf{E}}_{\beta}^{\infty}} = 2\mu_{\beta}^{\infty} \tilde{\mathbf{E}}_{\beta}^{\infty}, \quad \tilde{\mathbf{T}}^{\alpha} := \frac{\partial Y^{\alpha}}{\partial \tilde{\mathbf{E}}^{\alpha}} = -\frac{\partial Y^{\alpha}}{\partial \mathbf{E}^{\nu\alpha}} = 2\mu^{\alpha} (\tilde{\mathbf{E}}^{\alpha} - \mathbf{E}^{\nu\alpha}), \quad (2.56)$$

$$\tilde{\mathbf{Q}}_{\beta}^{\infty} := 2 \frac{\partial \tilde{\mathbf{E}}_{\beta}^{\infty}}{\partial \tilde{\mathbf{C}}}, \quad \tilde{\mathbf{Q}}^{\alpha} := 2 \frac{\partial \tilde{\mathbf{E}}^{\alpha}}{\partial \tilde{\mathbf{C}}}, \quad \text{and} \quad \mathbb{Q}^{\nu\alpha} := 2 \frac{\partial \mathbf{E}^{\nu\alpha}}{\partial \Gamma^{\alpha}}. \quad (2.57)$$

There are then M evolution equations, which are written as

$$\mathbf{Q}^{\alpha} = \mathbb{V}^{\alpha} : \left(\frac{1}{2} \frac{d\Gamma^{\alpha}}{dt} \right) \quad \text{or} \quad \frac{1}{2} \frac{d\Gamma^{\alpha}}{dt} = (\mathbb{V}^{\alpha})^{-1} : \mathbf{Q}^{\alpha}, \quad \alpha = 1, \dots, M. \quad (2.58)$$

The internal dissipation of the viscoelastic solid becomes

$$\mathcal{D} = \sum_{\alpha=1}^M \left(\frac{1}{2} \frac{d\Gamma^{\alpha}}{dt} \right) : \mathbb{V}^{\alpha} : \left(\frac{1}{2} \frac{d\Gamma^{\alpha}}{dt} \right) = \sum_{\alpha=1}^M \mathbf{Q}^{\alpha} : (\mathbb{V}^{\alpha})^{-1} : \mathbf{Q}^{\alpha}.$$

Now we show that the generalization made here does not disturb the result of Proposition 3.

Proposition 4. For the generalized model with the potential given in (2.53), the non-equilibrium stress $\mathbf{S}_{\text{iso}}^{\text{neq}}$ vanishes in the equilibrium state, that is,

$$\mathbf{S}_{\text{iso}}^{\text{neq}} \Big|_{\text{eq}} = \mathbf{0}.$$

Proof. According to Lemma 2, multiplying $\mathbb{Q}^{\nu\alpha-1}$ at both sides of (2.55)₃ leads to

$$\mathbf{Q}^{\alpha} : \mathbb{Q}^{\nu\alpha-1} = \tilde{\mathbf{T}}^{\alpha} : \mathbb{Q}^{\nu\alpha} : \mathbb{Q}^{\nu\alpha-1} = \tilde{\mathbf{T}}^{\alpha}.$$

It is known that $\mathbf{Q}^\alpha|_{\text{eq}} = \mathbf{O}$ due to the evolution Eq. (2.58). This implies $\tilde{\mathbf{T}}^\alpha|_{\text{eq}} = \mathbf{O}$, which leads to

$$\tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}|_{\text{eq}} = \sum_{\alpha=1}^M \tilde{\mathbf{T}}^\alpha|_{\text{eq}} : \tilde{\mathbf{Q}}^\alpha = \mathbf{O},$$

and hence one has $\mathbf{S}_{\text{iso}}^{\text{neq}}|_{\text{eq}} = J^{-\frac{2}{3}} \mathbb{P} : \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}|_{\text{eq}} = \mathbf{O}$. $\square$

We summarize the constitutive relations for finite viscoelastic materials discussed in Section 2.6.1 as follows.

Constitutive laws for finite viscoelasticity

Gibbs free energy:

$$\begin{aligned} G(\tilde{\mathbf{C}}, P, \mathbf{F}^1, \dots, \mathbf{F}^M) &= G_{\text{vol}}^\infty(P) + G_{\text{iso}}^\infty(\tilde{\mathbf{C}}) + Y(\tilde{\mathbf{C}}, \mathbf{F}^1, \dots, \mathbf{F}^M), \\ G_{\text{iso}}^\infty(\tilde{\mathbf{C}}) &= \sum_{\beta=1}^N G_{\text{iso}\beta}^\infty(\tilde{\mathbf{C}}), \quad G_{\text{iso}\beta}^\infty(\tilde{\mathbf{C}}) = \mu_\beta^\infty \left| \tilde{\mathbf{E}}_\beta^\infty(\tilde{\mathbf{C}}) \right|^2, \\ Y(\tilde{\mathbf{C}}, \mathbf{F}^1, \dots, \mathbf{F}^M) &= \sum_{\alpha=1}^M Y^\alpha(\tilde{\mathbf{C}}, \mathbf{F}^\alpha), \quad Y^\alpha(\tilde{\mathbf{C}}, \mathbf{F}^\alpha) = \mu^\alpha \left| \tilde{\mathbf{E}}^\alpha(\tilde{\mathbf{C}}) - \mathbf{E}^{\vee\alpha}(\mathbf{F}^\alpha) \right|^2. \end{aligned}$$

Projection tensors:

$$\tilde{\mathbf{Q}}_\beta^\infty = 2 \frac{\partial \tilde{\mathbf{E}}_\beta^\infty}{\partial \tilde{\mathbf{C}}}, \quad \mathbb{P} = \mathbb{I} - \frac{1}{3} \mathbf{C}^{-1} \otimes \mathbf{C}, \quad \tilde{\mathbf{Q}}^\alpha := 2 \frac{\partial \tilde{\mathbf{E}}^\alpha}{\partial \tilde{\mathbf{C}}}, \quad \mathbb{Q}^{\vee\alpha} = 2 \frac{\partial \mathbf{E}^{\vee\alpha}}{\partial \mathbf{F}^\alpha}.$$

Stress:

$$\begin{aligned} \mathbf{S} &= \mathbf{S}_{\text{iso}} + \mathbf{S}_{\text{vol}}, \quad \mathbf{S}_{\text{vol}} = -J P \mathbf{C}^{-1}, \quad \mathbf{S}_{\text{iso}} = \mathbf{S}_{\text{iso}}^\infty + \mathbf{S}_{\text{iso}}^{\text{neq}}, \\ \mathbf{S}_{\text{iso}}^\infty &= J^{-\frac{2}{3}} \mathbb{P} : \tilde{\mathbf{S}}_{\text{iso}}^\infty, \quad \tilde{\mathbf{S}}_{\text{iso}}^\infty = \sum_{\beta=1}^N \tilde{\mathbf{T}}_\beta^\infty : \tilde{\mathbf{Q}}_\beta^\infty, \quad \tilde{\mathbf{T}}_\beta^\infty = 2 \mu_\beta^\infty \tilde{\mathbf{E}}_\beta^\infty(\tilde{\mathbf{C}}), \\ \mathbf{S}_{\text{iso}}^{\text{neq}} &= J^{-\frac{2}{3}} \mathbb{P} : \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}}, \quad \tilde{\mathbf{S}}_{\text{iso}}^{\text{neq}} = \sum_{\alpha=1}^M \tilde{\mathbf{T}}^\alpha : \tilde{\mathbf{Q}}^\alpha, \quad \tilde{\mathbf{T}}^\alpha = 2 \mu^\alpha (\tilde{\mathbf{E}}^\alpha(\tilde{\mathbf{C}}) - \mathbf{E}^{\vee\alpha}(\mathbf{F}^\alpha)). \end{aligned}$$

Constitutive relations for $\mathbf{Q}^\alpha$:

$$\mathbf{Q}^\alpha = \tilde{\mathbf{T}}^\alpha : \mathbb{Q}^{\vee\alpha}.$$

Dissipation:

$$\mathbf{D} = \sum_{\alpha=1}^M \left(\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} \right) : \mathbb{V}^\alpha : \left(\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} \right) = \sum_{\alpha=1}^M \mathbf{Q}^\alpha : (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha.$$

Evolution equations:

$$\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} = (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha.$$

3. Numerical aspects

In this section, we design numerical methods for the viscoelasticity model based on the Gibbs free energy. The numerical design involves the spatiotemporal discretization of the balance equations and the integration of the constitutive equation. The former follows the numerical scheme introduced in Part I, and our discussion focuses on the constitutive integration as well as the consistent linearization of the system.

3.1. Evolution equation integration

The algorithmic procedure for evaluating $\mathbf{Q}^\alpha$ is based on the evolution equations for $\mathbf{F}^\alpha$, which reads

$$\frac{1}{2} \frac{d\mathbf{F}^\alpha}{dt} = (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha. \quad (3.1)$$

We assume material is Newtonian, meaning $\mathbb{V}^\alpha$ is a constant tensor. Integrating the nonlinear evolution equations entails a local Newton–Raphson iterative procedure at each quadrature point. During the linearization process, it is necessary to invoke a set of rank-four tensors defined as follows,

$$\mathbb{K}^\alpha := 2 \frac{\partial \mathbf{Q}^\alpha}{\partial \mathbf{F}^\alpha} = 2 \frac{\partial (\tilde{\mathbf{T}}^\alpha : \mathbb{Q}^{\vee\alpha})}{\partial \mathbf{F}^\alpha} = \mathbb{Q}^{\vee\alpha T} : 2 \frac{\partial \tilde{\mathbf{T}}^\alpha}{\partial \mathbf{F}^\alpha} + \tilde{\mathbf{T}}^\alpha : 2 \frac{\partial \mathbb{Q}^{\vee\alpha}}{\partial \mathbf{F}^\alpha}$$

$$\begin{aligned}
&= -2\mu^\alpha \mathbb{Q}^{\vee \alpha T} : \mathbb{Q}^{\vee \alpha} + \tilde{\mathbf{T}}^\alpha : \mathbb{L}^{\vee \alpha}, \\
\mathbb{J}^\alpha &:= 2 \frac{\partial \mathbf{Q}^\alpha}{\partial \tilde{\mathbf{C}}} = 2 \frac{\partial(\tilde{\mathbf{T}}^\alpha : \mathbb{Q}^{\vee \alpha})}{\partial \tilde{\mathbf{C}}} = \mathbb{Q}^{\vee \alpha T} : 2 \frac{\partial \tilde{\mathbf{T}}^\alpha}{\partial \tilde{\mathbf{C}}} + \tilde{\mathbf{T}}^\alpha : 2 \frac{\partial \mathbb{Q}^{\vee \alpha}}{\partial \tilde{\mathbf{C}}} \\
&= 2\mu^\alpha \mathbb{Q}^{\vee \alpha T} : \tilde{\mathbb{Q}}^\alpha,
\end{aligned} \tag{3.2}$$

with

$$\mathbb{L}^{\vee \alpha} := 4 \frac{\partial^2 \mathbf{E}^{\vee \alpha}}{\partial \mathbf{F}^\alpha \partial \mathbf{F}^\alpha} = 2 \frac{\partial \mathbb{Q}^{\vee \alpha}}{\partial \mathbf{F}^\alpha}. \tag{3.4}$$

In the above, we have used the fact that $2\partial \tilde{\mathbf{T}}^\alpha / \partial \mathbf{F}^\alpha = -2\mu^\alpha \mathbb{Q}^{\vee \alpha}$ and $2\partial \tilde{\mathbf{T}}^\alpha / \partial \tilde{\mathbf{C}} = 2\mu^\alpha \tilde{\mathbb{Q}}^\alpha$.

Let the time interval $(0, T)$ be divided into n_{ts} subintervals of size $\Delta t_n := t_{n+1} - t_n$ delimited by a discrete time vector $\{t_n\}_{n=0}^{n_{\text{ts}}}$. We integrate the evolution Eq. (3.1) by the mid-point rule, leading to the following discrete evolution equations,

$$\frac{1}{2} \frac{\mathbf{F}_{n+1}^\alpha - \mathbf{F}_n^\alpha}{\Delta t_n} = (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}}^\alpha \right), \tag{3.5}$$

with

$$(\bullet)_{n+\frac{1}{2}} := \frac{1}{2} ((\bullet)_n + (\bullet)_{n+1}).$$

In the above, $(\bullet)_n$ represents the approximation of the quantity $(\bullet)$ at the time instance t_n . We summarize the local Newton–Raphson iteration procedure at each quadrature point for determining $\mathbf{F}_{n+1}^\alpha$ as follows.

Local Newton–Raphson iteration

Predictor stage: Set

$$\mathbf{F}_{n+1(0)}^\alpha = \mathbf{F}_n^\alpha,$$

for $\alpha = 1, \dots, M$.

Multi-Corrector stage: Repeat the following steps for $\alpha = 1, \dots, M$.

1. Construct the local residual

$$\mathbf{r}_{n+1(i)}^\alpha := \mathbf{F}_{n+1(i)}^\alpha - \mathbf{F}_n^\alpha - 2\Delta t_n (\mathbb{V}^\alpha)^{-1} : \mathbf{Q}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}(i)}^\alpha \right),$$

with

$$\mathbf{F}_{n+\frac{1}{2}(i)}^\alpha := \frac{1}{2} \left(\mathbf{F}_n^\alpha + \mathbf{F}_{n+1(i)}^\alpha \right).$$

If one of the stopping criteria

$$\left| \frac{\mathbf{r}_{n+1(i)}^\alpha}{\mathbf{r}_{n+1(0)}^\alpha} \right| \leq \text{tol}_r, \quad \left| \mathbf{r}_{n+1(i)}^\alpha \right| \leq \text{tol}_a, \quad \text{and} \quad i = i_{\max}$$

is satisfied, set $\mathbf{F}_{n+1}^\alpha = \mathbf{F}_{n+1(i)}^\alpha$ and exit the multi-corrector stage. Otherwise, continue to Step 2.

2. Linearize the local residual

$$\left. \frac{\partial \mathbf{r}_{n+1}^\alpha}{\partial \mathbf{F}^\alpha} \right|_{(i)} = \mathbb{S}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}(i)}^\alpha \right),$$

with

$$\mathbb{S}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}(i)}^\alpha \right) := \mathbb{I} - \frac{\Delta t_n}{2} (\mathbb{V}^\alpha)^{-1} : \mathbb{K}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}(i)}^\alpha \right). \tag{3.6}$$

3. Solve the incremental solution

$$\mathbb{S}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}(i)}^\alpha \right) : \Delta \mathbf{F}_{n+1(i)}^\alpha = -\mathbf{r}_{n+1(i)}^\alpha.$$

4. Update the discrete internal state variable as

$$\mathbf{F}_{n+1(i+1)}^\alpha = \mathbf{F}_{n+1(i)}^\alpha + \Delta \mathbf{F}_{n+1(i)}^\alpha.$$

Remark 13. In the multiplicative theory of viscoelasticity, it is common to assume the inelastic flow is volume-preserving [40,109]. This property is inherited to the discrete level through the use of the exponential map integrator [16,30]. If we want to analogously

enforce the isochoric condition for the internal state variables $\{\Gamma^\alpha\}$, a volume-preserving constitutive integration is worth of pursuing and the scheme introduced in [110] shows promise in this regard.

3.2. Consistent linearization

Consistent linearization is the cornerstone in the algorithm of the Newton–Raphson type [11]. In the following, we provide the closed-form formula for the elasticity tensor associated with the model outlined in Section 2.6.1. We introduce the isochoric elasticity tensor as

$$\mathbb{C}_{\text{iso } n+1} := 2 \frac{\partial \mathcal{S}_{\text{iso } n+1}}{\partial \mathbf{C}_{n+1}} = \mathbb{C}_{\text{iso } n+1}^\infty + \mathbb{C}_{\text{iso } n+1}^{\text{neq}}, \quad \text{with} \quad \mathbb{C}_{\text{iso } n+1}^\infty := 2 \frac{\partial \mathcal{S}_{\text{iso } n+1}^\infty}{\partial \mathbf{C}_{n+1}} \quad \text{and} \quad \mathbb{C}_{\text{iso } n+1}^{\text{neq}} := 2 \frac{\partial \mathcal{S}_{\text{iso } n+1}^{\text{neq}}}{\partial \mathbf{C}_{n+1}}, \quad (3.7)$$

due to the fact that $\mathcal{S}_{\text{iso } n+1} = \mathcal{S}_{\text{iso } n+1}^\infty + \mathcal{S}_{\text{iso } n+1}^{\text{neq}}$. With the representation of $\mathcal{S}_{\text{iso } n+1}^\infty$ given in (2.49)₂, the equilibrium part of $\mathbb{C}_{\text{iso } n+1}$ can be represented as

$$\mathbb{C}_{\text{iso } n+1}^\infty = \mathbb{P}_{n+1} : \tilde{\mathbb{C}}_{\text{iso } n+1}^\infty : \mathbb{P}_{n+1} + \frac{2}{3} \text{Tr} \left(J_{n+1}^{-\frac{2}{3}} \tilde{\mathcal{S}}_{\text{iso } n+1}^\infty \right) \tilde{\mathbb{P}}_{n+1} - \frac{2}{3} (\mathbf{C}_{n+1}^{-1} \otimes \mathcal{S}_{\text{iso } n+1}^\infty + \mathcal{S}_{\text{iso } n+1}^\infty \otimes \mathbf{C}_{n+1}^{-1}), \quad (3.8)$$

with the following notations,

$$\tilde{\mathbb{C}}_{\text{iso } n+1}^\infty := 2 J_{n+1}^{-\frac{4}{3}} \frac{\partial \tilde{\mathcal{S}}_{\text{iso } n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1}}, \quad \text{Tr}(\cdot) := (\cdot) : \mathbf{C}_{n+1}, \quad \tilde{\mathbb{P}}_{n+1} := \mathbf{C}_{n+1}^{-1} \odot \mathbf{C}_{n+1}^{-1} - \frac{1}{3} \mathbf{C}_{n+1}^{-1} \otimes \mathbf{C}_{n+1}^{-1}. \quad (3.9)$$

Readers may refer to [65, p. 255] for a detailed derivation of the formula (3.8). Moreover, due to the constitutive model of $\tilde{\mathcal{S}}_{\text{iso } n+1}^\infty$ given in (2.55)₁, we have the following explicit formula for $\tilde{\mathbb{C}}_{\text{iso } n+1}^\infty$,

$$\begin{aligned} \tilde{\mathbb{C}}_{\text{iso } n+1}^\infty &= 2 J_{n+1}^{-\frac{4}{3}} \frac{\partial \tilde{\mathcal{S}}_{\text{iso } n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1}} = J_{n+1}^{-\frac{4}{3}} \sum_{\beta=1}^N 2 \frac{\partial (\tilde{\mathbf{T}}_{\beta n+1}^\infty : \tilde{\mathbb{Q}}_{\beta n+1}^\infty)}{\partial \tilde{\mathbf{C}}_{n+1}} \\ &= J_{n+1}^{-\frac{4}{3}} \sum_{\beta=1}^N \left(\tilde{\mathbb{Q}}_{\beta n+1}^\infty T : 2 \frac{\partial \tilde{\mathbf{T}}_{\beta n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1}} + \tilde{\mathbf{T}}_{\beta n+1}^\infty : 2 \frac{\partial \tilde{\mathbb{Q}}_{\beta n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1}} \right) \\ &= J_{n+1}^{-\frac{4}{3}} \sum_{\beta=1}^N \left(2 \mu_\beta^\infty \tilde{\mathbb{Q}}_{\beta n+1}^\infty T : \tilde{\mathbb{Q}}_{\beta n+1}^\infty + \tilde{\mathbf{T}}_{\beta n+1}^\infty : \tilde{\mathbb{L}}_{\beta n+1}^\infty \right), \end{aligned} \quad (3.10)$$

with

$$\tilde{\mathbb{L}}_{\beta n+1}^\infty := 4 \frac{\partial^2 \tilde{\mathbf{E}}_{\beta n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1} \partial \tilde{\mathbf{C}}_{n+1}} = 2 \frac{\partial \tilde{\mathbb{Q}}_{\beta n+1}^\infty}{\partial \tilde{\mathbf{C}}_{n+1}}. \quad (3.11)$$

In the above, we have used the fact that $2 \partial \tilde{\mathbf{T}}_{\beta n+1}^\infty / \partial \tilde{\mathbf{C}}_{n+1} = 2 \mu_\beta^\infty \tilde{\mathbb{Q}}_{\beta n+1}^\infty$.

The derivation and final expression of $\mathbb{C}_{\text{iso } n+1}^{\text{neq}}$ in (3.7) are analogous to those of $\mathbb{C}_{\text{iso } n+1}^\infty$ above. We give the final closed-form formula of $\mathbb{C}_{\text{iso } n+1}^{\text{neq}}$ as

$$\mathbb{C}_{\text{iso } n+1}^{\text{neq}} = \mathbb{P}_{n+1} : \tilde{\mathbb{C}}_{\text{iso } n+1}^{\text{neq}} : \mathbb{P}_{n+1} + \frac{2}{3} \text{Tr} \left(J_{n+1}^{-\frac{2}{3}} \tilde{\mathcal{S}}_{\text{iso } n+1}^{\text{neq}} \right) \tilde{\mathbb{P}}_{n+1} - \frac{2}{3} (\mathbf{C}_{n+1}^{-1} \otimes \mathcal{S}_{\text{iso } n+1}^{\text{neq}} + \mathcal{S}_{\text{iso } n+1}^{\text{neq}} \otimes \mathbf{C}_{n+1}^{-1}), \quad (3.12)$$

with

$$\begin{aligned} \tilde{\mathbb{C}}_{\text{iso } n+1}^{\text{neq}} &:= 2 J_{n+1}^{-\frac{4}{3}} \frac{\partial \tilde{\mathcal{S}}_{\text{iso } n+1}^{\text{neq}}}{\partial \tilde{\mathbf{C}}_{n+1}} = J_{n+1}^{-\frac{4}{3}} \sum_{\alpha=1}^M 2 \frac{\partial (\tilde{\mathbf{T}}_{n+1}^\alpha : \tilde{\mathbb{Q}}_{n+1}^\alpha)}{\partial \tilde{\mathbf{C}}_{n+1}} \\ &= J_{n+1}^{-\frac{4}{3}} \sum_{\alpha=1}^M \left(\tilde{\mathbb{Q}}_{n+1}^\alpha T : 2 \frac{\partial \tilde{\mathbf{T}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} + \tilde{\mathbf{T}}_{n+1}^\alpha : 2 \frac{\partial \tilde{\mathbb{Q}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} \right) \\ &= J_{n+1}^{-\frac{4}{3}} \sum_{\alpha=1}^M \left(4 \mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \frac{\partial \tilde{\mathbf{E}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} + \tilde{\mathbf{T}}_{n+1}^\alpha : 2 \frac{\partial \tilde{\mathbb{Q}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} - 4 \mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \frac{\partial \mathbf{E}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} \right) \\ &= J_{n+1}^{-\frac{4}{3}} \sum_{\alpha=1}^M \left(2 \mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \tilde{\mathbb{Q}}_{n+1}^\alpha + \tilde{\mathbf{T}}_{n+1}^\alpha : \tilde{\mathbb{L}}_{n+1}^\alpha - 2 \mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \mathbb{Q}_{n+1}^\alpha : \frac{\partial \Gamma_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} \right), \end{aligned} \quad (3.13)$$

and

$$\tilde{\mathbb{L}}_{n+1}^\alpha := 4 \frac{\partial^2 \tilde{\mathbf{E}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1} \partial \tilde{\mathbf{C}}_{n+1}} = 2 \frac{\partial \tilde{\mathbb{Q}}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}}. \quad (3.14)$$

We may obtain $\partial \Gamma_{n+1}^\alpha / \partial \tilde{\mathbf{C}}_{n+1}$ by taking derivative of the discrete evolution Eq. (3.5) with respect to $\tilde{\mathbf{C}}_{n+1}$, which leads to

$$\frac{\partial \Gamma_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} = 2 \Delta t (\nabla^\alpha)^{-1} : \left[\frac{\partial \mathbf{Q}^\alpha}{\partial \tilde{\mathbf{C}}_{n+\frac{1}{2}}} : \frac{\partial \tilde{\mathbf{C}}_{n+\frac{1}{2}}}{\partial \tilde{\mathbf{C}}_{n+1}} + \frac{\partial \mathbf{Q}^\alpha}{\partial \tilde{\mathbf{C}}_{n+\frac{1}{2}}} : \frac{\partial \Gamma_{n+\frac{1}{2}}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} \right]$$

$$= \frac{\Delta t}{2} (\mathbb{V}^\alpha)^{-1} : \left[\mathbb{J}^\alpha (\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}}^\alpha) + \mathbb{K}^\alpha (\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}}^\alpha) : \frac{\partial \mathbf{F}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} \right].$$

The above relation can be reorganized as

$$\mathbb{S}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}}^\alpha \right) : \frac{\partial \mathbf{F}_{n+1}^\alpha}{\partial \tilde{\mathbf{C}}_{n+1}} = \frac{\Delta t}{2} (\mathbb{V}^\alpha)^{-1} : \mathbb{J}^\alpha \left(\tilde{\mathbf{C}}_{n+\frac{1}{2}}, \mathbf{F}_{n+\frac{1}{2}}^\alpha \right), \quad (3.15)$$

in which we have invoked the definition Eq. (3.6). The rank-four tensor $\partial \mathbf{F}_{n+1}^\alpha / \partial \tilde{\mathbf{C}}_{n+1}$ that arises in (3.13) is determined by solving the above linear Eq. (3.15).

Remark 14. The elasticity tensor $\mathbb{C}_{\text{iso}}$ is represented through the tensorial operations of the stresses and projection tensors, allowing modular implementation for different strains. To complete the definition of the consistent tangent matrix, one also needs the geometric stiffness tensor, which can be calculated by the current stress state [27, p. 534]. Interestingly, a closed-form of the geometric stiffness tensor is derived for stretch-based models [111]. The analytic form facilitates the design of tangent stabilization, which has been demonstrated to significantly enhance the robustness of the Newton–Raphson type solution strategy [112].

3.3. Modular implementation

Given that the material model under consideration relies on the hyperelasticity of Hill's class, the implementation of the constitutive relations can be performed in a modular approach. It follows the algorithmic steps summarized by Miehe in additive plasticity [49,51,113] and entails three essential steps: the geometric preprocessing, constitutive model evaluation in the generalized strain space, and geometric postprocessing. In the geometric preprocessing step, the strains and projection tensors are evaluated based on the deformation tensor and internal state variables at each quadrature point.

Geometric preprocessing: Given the deformation gradient $\mathbf{F}_{n+1}$ and the internal state variables $\{\mathbf{F}_n^\alpha\}$, perform the following procedures.

1. Calculate the unimodular deformation tensor $\tilde{\mathbf{C}}_{n+1}$.
2. Perform the spectral decomposition on the deformation tensor to obtain the principal stretches $\{\tilde{\lambda}_{a\,n+1}\}$ and Lagrangian principal directions $\{\mathbf{N}_{a\,n+1}\}$ for $a = 1, 2, 3$.
3. Based on the scale functions E_β^∞ , calculate the strain

$$\tilde{\mathbf{E}}_{\beta\,n+1}^\infty = \sum_{a=1}^3 E_\beta^\infty(\tilde{\lambda}_{a\,n+1}) \mathbf{M}_{a\,n+1}$$

and the related projection tensors $\tilde{\mathbb{Q}}_{\beta\,n+1}^\infty$ and $\tilde{\mathbb{L}}_{\beta\,n+1}^\infty$ for $\beta = 1, \dots, N$.

4. Based on the scale functions E^α , calculate the strain

$$\tilde{\mathbf{E}}_{n+1}^\alpha = \sum_{a=1}^3 E^\alpha(\tilde{\lambda}_{a\,n+1}) \mathbf{M}_{a\,n+1}$$

and the related projection tensors $\tilde{\mathbb{Q}}_{n+1}^\alpha$ and $\tilde{\mathbb{L}}_{n+1}^\alpha$ for $\alpha = 1, \dots, M$.

5. Obtain $\mathbf{F}_{n+1}^\alpha$ for $\alpha = 1, \dots, M$ by invoking the local Newton–Raphson iteration outlined in Section 3.1.
6. Obtain the rank-four tensor $\partial \mathbf{F}_{n+1}^\alpha / \partial \tilde{\mathbf{C}}_{n+1}$ by solving the linear Eq. (3.15).
7. Perform the spectral decompositions on $\{\mathbf{F}_{n+1}^\alpha\}$ to obtain $\{\mathbf{F}_{a\,n+1}^\alpha\}$ and $\{\tilde{\mathbf{N}}_{a\,n+1}^\alpha\}$ for $\alpha = 1, \dots, M$ and $a = 1, 2, 3$.
8. Based on the scale functions $E^{\vee\alpha}$, calculate the strain

$$\mathbf{E}_{n+1}^{\vee\alpha} = \sum_{a=1}^3 E^{\vee\alpha}(\mathbf{F}_{a\,n+1}^\alpha) \tilde{\mathbf{M}}_{a\,n+1}$$

for $\alpha = 1, \dots, M$.

Remark 15. In the above preprocessing procedures, we invoked the spectral decomposition algorithm developed by Scherzinger and Dohrmann [92]. Its accuracy is comparable to that of LAPACK [114], and its robustness is superior to the conventional Cardano's formula when there are nearly identical eigenvalues [93]. Additionally, the projection tensors are computed following the formulas (2.8) and (2.11) using the corresponding eigenvalues and eigenvectors.

Moving to the second step, we conduct the constitutive evaluation in the generalized strain space. As a main feature of Hill's hyperelasticity, the conjugate stress-like tensors depend on the generalized strains in a linear fashion and can be straightforwardly evaluated.

Constitutive model evaluation: Given the generalized strains, perform the following procedures to obtain their work-conjugate pairs as well as the fictitious elasticity tensors.

1. Evaluate the stresses $\tilde{\mathbf{T}}_{\beta\,n+1}^\infty = 2\mu_\beta^\infty \tilde{\mathbf{E}}_{\beta\,n+1}^\infty$ for $\beta = 1, \dots, N$.

2. Evaluate the stresses $\tilde{T}_{n+1}^\alpha = 2\mu^\alpha (\tilde{E}_{n+1}^\alpha - E_{n+1}^{\nu\alpha})$ for $\alpha = 1, \dots, M$.
3. Evaluate the fictitious elasticity tensors for the equilibrium part as

$$\tilde{\mathbb{C}}_{\text{iso } n+1}^\infty = J_{n+1}^{-\frac{4}{3}} \sum_{\beta=1}^N \left(2\mu_\beta^\infty \tilde{\mathbb{Q}}_{\beta n+1}^\infty T : \tilde{\mathbb{Q}}_{\beta n+1}^\infty + \tilde{T}_{\beta n+1}^\infty : \tilde{\mathbb{L}}_{\beta n+1}^\infty \right).$$

4. Evaluate the fictitious elasticity tensors for the non-equilibrium part as

$$\tilde{\mathbb{C}}_{\text{iso } n+1}^{\text{neq}} := J_{n+1}^{-\frac{4}{3}} \sum_{\alpha=1}^M \left(2\mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \tilde{\mathbb{Q}}_{n+1}^\alpha + \tilde{T}_{n+1}^\alpha : \tilde{\mathbb{L}}_{n+1}^\alpha - 2\mu^\alpha \tilde{\mathbb{Q}}_{n+1}^\alpha T : \mathbb{Q}_{n+1}^{\nu\alpha} : \frac{\partial \Gamma_{n+1}^\alpha}{\partial \tilde{\mathbb{C}}_{n+1}} \right).$$

In the last step, known as geometric postprocessing, the Piola–Kirchhoff stresses and elasticity tensors are obtained with the aid of the projectors.

Geometric postprocessing: Obtain the second Piola–Kirchhoff stresses and the elasticity tensor through the following procedures.

1. Obtain the equilibrium part of the fictitious stress as

$$\tilde{\mathbb{S}}_{\text{iso } n+1}^\infty = \sum_{\beta=1}^N \tilde{T}_{\beta n+1}^\infty : \tilde{\mathbb{Q}}_{\beta n+1}^\infty.$$

2. Obtain the non-equilibrium part of the fictitious stress as

$$\tilde{\mathbb{S}}_{\text{iso } n+1}^{\text{neq}} = \sum_{\alpha=1}^M \tilde{T}_{n+1}^\alpha : \tilde{\mathbb{Q}}_{n+1}^\alpha.$$

3. Project the fictitious stresses to the second Piola–Kirchhoff stress as

$$\mathbb{S}_{\text{iso } n+1}^\infty = J_{n+1}^{-\frac{2}{3}} \mathbb{P}_{n+1} : \tilde{\mathbb{S}}_{\text{iso } n+1}^\infty, \quad \mathbb{S}_{\text{iso } n+1}^{\text{neq}} = J_{n+1}^{-\frac{2}{3}} \mathbb{P}_{n+1} : \tilde{\mathbb{S}}_{\text{iso } n+1}^{\text{neq}}, \quad \mathbb{S}_{\text{iso } n+1} = \mathbb{S}_{\text{iso } n+1}^\infty + \mathbb{S}_{\text{iso } n+1}^{\text{neq}}.$$

4. Obtain the equilibrium part of the elasticity tensor as

$$\begin{aligned} \mathbb{C}_{\text{iso } n+1}^\infty = \mathbb{P}_{n+1} : \tilde{\mathbb{C}}_{\text{iso } n+1}^\infty : \mathbb{P}_{n+1}^T + \frac{2}{3} \text{Tr} \left(J_{n+1}^{-\frac{2}{3}} \tilde{\mathbb{S}}_{\text{iso } n+1}^\infty \right) \tilde{\mathbb{P}}_{n+1} \\ - \frac{2}{3} \left(C_{n+1}^{-1} \otimes \mathbb{S}_{\text{iso } n+1}^\infty + \mathbb{S}_{\text{iso } n+1}^\infty \otimes C_{n+1}^{-1} \right). \end{aligned}$$

5. Obtain the non-equilibrium part of the elasticity tensor as

$$\begin{aligned} \mathbb{C}_{\text{iso } n+1}^{\text{neq}} = \mathbb{P}_{n+1} : \tilde{\mathbb{C}}_{\text{iso } n+1}^{\text{neq}} : \mathbb{P}_{n+1}^T + \frac{2}{3} \text{Tr} \left(J_{n+1}^{-\frac{2}{3}} \tilde{\mathbb{S}}_{\text{iso } n+1}^{\text{neq}} \right) \tilde{\mathbb{P}}_{n+1} \\ - \frac{2}{3} \left(C_{n+1}^{-1} \otimes \mathbb{S}_{\text{iso } n+1}^{\text{neq}} + \mathbb{S}_{\text{iso } n+1}^{\text{neq}} \otimes C_{n+1}^{-1} \right). \end{aligned}$$

6. Obtain the isochoric part of the elasticity tensor as

$$\mathbb{C}_{\text{iso } n+1} = \mathbb{C}_{\text{iso } n+1}^\infty + \mathbb{C}_{\text{iso } n+1}^{\text{neq}}.$$

Through the above three modular procedures, we may obtain the stress and elasticity tensor that are necessary for the assembly of the residual vector and consistent tangent matrix in implicit analysis. We mention that the implementation of the algorithm relies heavily on efficient calculations of tensorial operations, in which the tensor symmetry can be exploited to save memory and computational costs.

4. Examples

In this section, we investigate the performance of the proposed model. The boundary of the domain $\Gamma_X := \partial\Omega_X$ admits non-overlapping subdivisions as $\Gamma_X = \Gamma_X^G \cup \Gamma_X^H$. The functions $\mathbf{G}$ and $\mathbf{H}$ are the boundary displacement and traction prescribed on Γ_X^G and Γ_X^H , respectively. In more involved scenarios, we may prescribe, for example, the normal component of the displacement and the tangential components of the traction on a boundary region. That type of boundary condition will be specified with the aid of Cartesian components of the functions $\mathbf{G}$ and $\mathbf{H}$. The problem is discretized by an inf-sup stable element pair [102]. The discrete pressure space is generated by the NURBS basis functions with degree p and highest possible continuity. The discrete velocity space is constructed by p -refinement to raise the degree to $p+1$ while maintaining the continuity. The time integration is based on the generalized- α scheme [98,115,116], which is parameterized by a single parameter ρ_∞ , the spectral radius of the amplification matrix at the highest mode. The following choices are made in the numerical investigations, unless otherwise specified.

1. The meter-kilogram-second system of units is used;

Table 2
Creep test: problem setting.

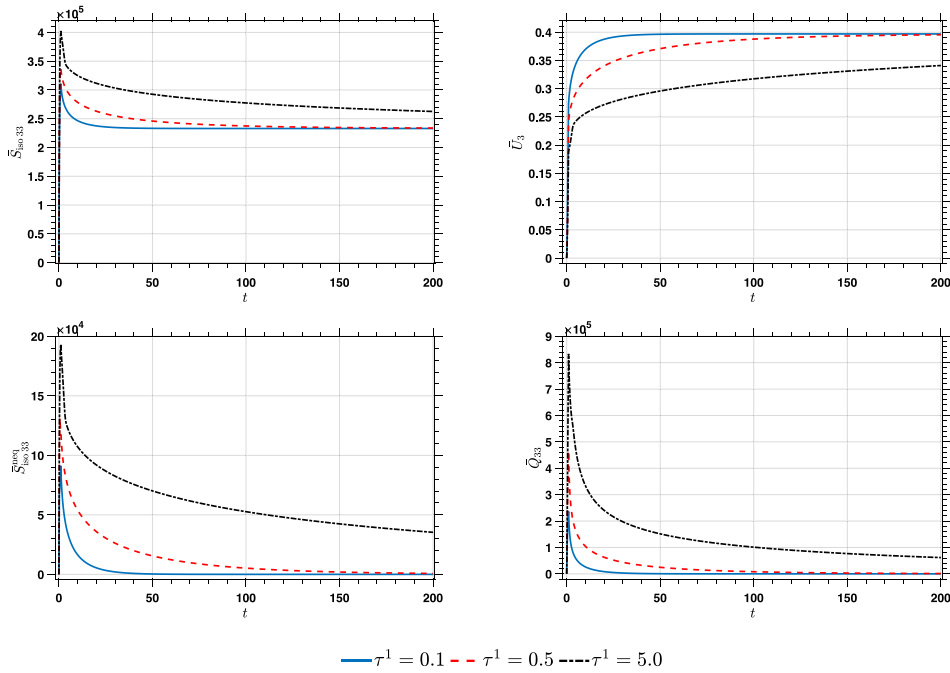
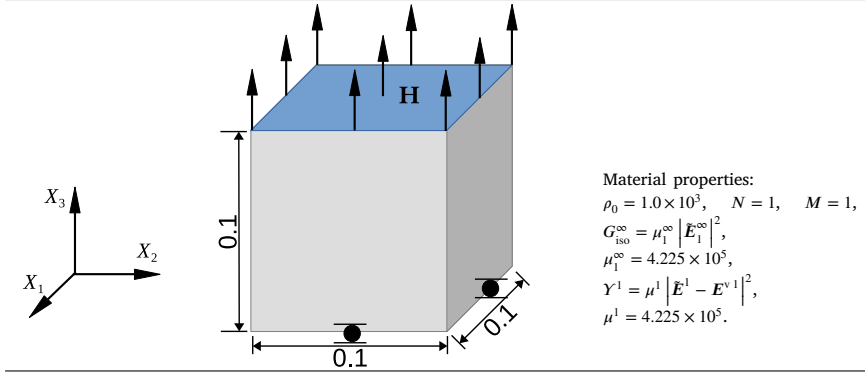


Fig. 7. Creep test: the effects of the relaxation time. The strain is given by the parameters $m = 1.2$ and $n = 1.4$.

2. The two-parameter Curnier–Rakotomanana strain family is utilized to characterize the hyperelastic material behavior within Hill’s framework;
3. The viscosity tensor is isotropic with the deviatoric and volumetric viscosities being identical (i.e., $\mathbb{V}^\alpha = 2\eta^\alpha \mathbb{I}$);
4. The relaxation times are defined as $\tau^\alpha := \eta^\alpha / \mu^\alpha$;
5. We choose $\rho_\infty = 0.0$ in the generalized- α scheme;
6. We use $p + 2$ Gaussian quadrature points in each direction;
7. We use parameters $\text{tol}_r = 10^{-10}$, $\text{tol}_a = 10^{-10}$, $i_{\max} = 5$ in the local Newton–Raphson iteration;
8. We use the relative tolerance $\text{tol}_R = 10^{-10}$, absolute tolerance $\text{tol}_A = 10^{-10}$, and maximum number of iterations $i_{\max} = 10$ as the stopping criteria in the global Newton–Raphson iteration.

4.1. Creep test

We investigate the creep behavior with the problem setting summarized in Table 2. The strains $\tilde{\mathbf{E}}_1^\infty$, $\tilde{\mathbf{E}}^1$, and $\mathbf{E}^{v1}$ are defined by the same two parameters m and n , which will be specified momentarily. The problem is integrated up to $T = 200$. On the bottom boundary, the tangential components of the traction are zero (i.e., $H_1 = H_2 = 0$), and the normal displacement is fixed (i.e., $G_3 = 0$).

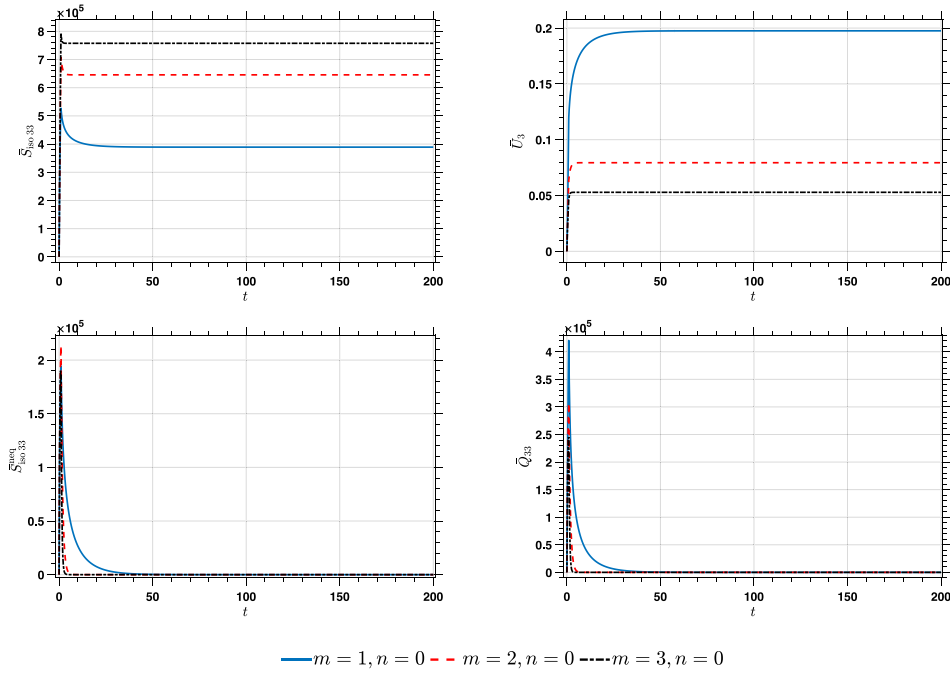


Fig. 8. Creep test: the effects of the different strains. The relaxation time is 0.5.

Table 3

Creep test: convergence of the Newton–Raphson iteration.

Iteration	l_2 -norm of the residual vector		
	$m = 1, n = 0$ 90th time step	$m = 2, n = 0$ 90th time step	$m = 3, n = 0$ 60th time step
1	5.778×10^0	3.422×10^{-1}	1.626×10^{-1}
2	2.850×10^{-5}	1.164×10^{-6}	1.001×10^{-6}
3	8.458×10^{-11}	1.084×10^{-11}	3.624×10^{-12}

On the top boundary Γ_{top} , the traction of the form $\mathbf{H} = [0, 0, H_3(t)]^T$ is imposed with

$$H_3(t) = \begin{cases} 1.737 \times 10^6 t, & t \leq 1.0, \\ 1.737 \times 10^6, & 1.0 < t \leq T. \end{cases}$$

On the rest boundary surfaces, stress-free boundary conditions are applied. In this example, we monitor the following four averaged quantities on the top surface,

$$\bar{S}_{\text{iso } 33} := \frac{\int_{\Gamma_{\text{top}}} S_{\text{iso } 33} d\Gamma_X}{\int_{\Gamma_{\text{top}}} d\Gamma_X}, \quad \bar{U}_3 := \frac{\int_{\Gamma_{\text{top}}} U_3 d\Gamma_X}{\int_{\Gamma_{\text{top}}} d\Gamma_X}, \quad \bar{S}_{\text{iso } 33}^{\text{neq}} := \frac{\int_{\Gamma_{\text{top}}} S_{\text{iso } 33}^{\text{neq}} d\Gamma_X}{\int_{\Gamma_{\text{top}}} d\Gamma_X}, \quad \bar{Q}_{33} := \frac{\int_{\Gamma_{\text{top}}} Q_{33} d\Gamma_X}{\int_{\Gamma_{\text{top}}} d\Gamma_X}.$$

Prior to our investigation, we have adopted two meshes to guarantee the mesh independence of the reported results. The following results are based on a time step size of $\Delta t_n = 0.05$ and a mesh of $5 \times 5 \times 5$ elements with the polynomial degree $p = 2$. In Fig. 7, we analyze the model behavior with three different relaxation times. It can be observed that the averaged stress $\bar{S}_{\text{iso } 33}$ and the averaged displacement $\bar{U}_3$ approach their equilibrium values at the end of the simulation, for the cases with smaller values of τ^1 . The corresponding values of $\bar{Q}_{33}$ and $\bar{S}_{\text{iso } 33}^{\text{neq}}$ both tend towards zero, confirming the relaxation property. With larger values of the relaxation time, the model requires a longer period of time to achieve the equilibrium state, and the monitored quantities remain far from their equilibrium values. In Fig. 8, we examine the influence of the generalized strains by considering different choices of m and n in the scale function of the Curnier–Rakotomanana family. In particular, we consider the case of $m = 2$ and $n = 0$, which corresponds to the finite deformation linear model [1]. It is observed that different strains have a significant impact on the values of $\bar{S}_{\text{iso } 33}$ and $\bar{U}_3$ at the equilibrium state. In the meantime, the averaged stresses $\bar{S}_{\text{iso } 33}^{\text{neq}}$ and $\bar{Q}_{33}$ vanish with different rates depending on the strains. We also illustrate the convergence rate of the Newton–Raphson iteration in Table 3 at randomly selected time steps, where the l_2 -norms of the residual vectors are recorded. Only three iterations are needed to attain convergence for all three models.

Table 4
Shear test: problem setting.

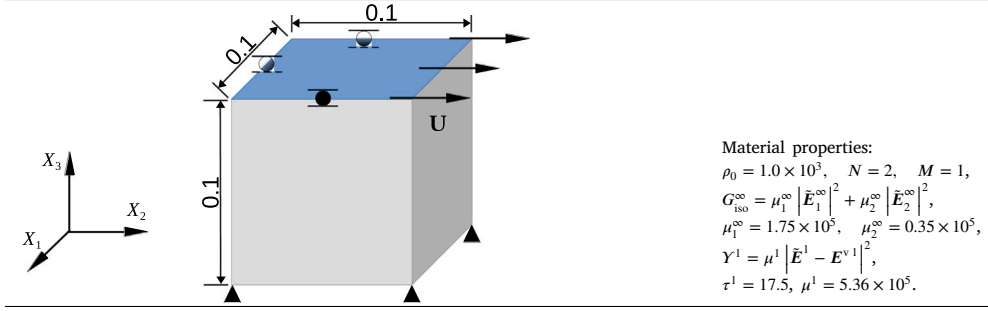


Table 5
Convergence of Newton–Raphson iteration in the 35th time step.

Iteration	L_2 -norm of the residual vector		
	$m = 0, n = 0$	$m = 1, n = 0$	$m = 2, n = 0$
1	3.235×10^{-3}	3.910×10^{-3}	9.223×10^{-3}
2	2.697×10^{-6}	3.061×10^{-6}	3.706×10^{-6}
3	5.661×10^{-11}	5.706×10^{-11}	9.219×10^{-11}

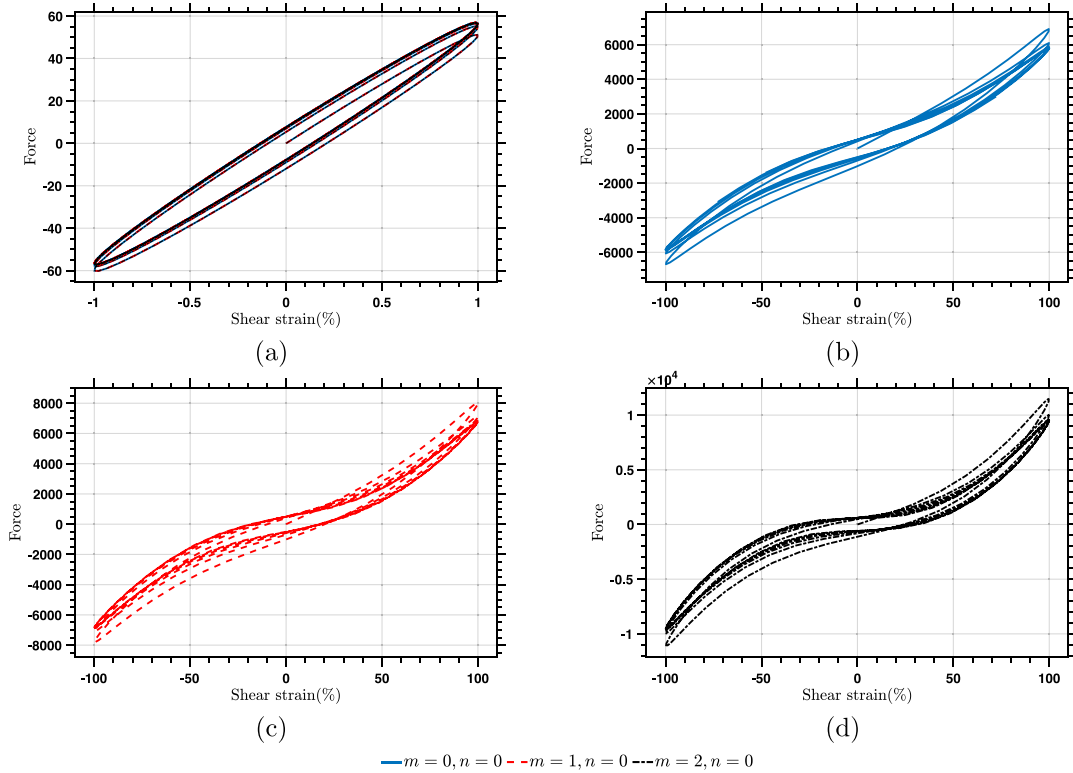


Fig. 9. Shear test: the hysteresis loops of different strains. The case with $U_0/L_0 = 1\%$ is illustrated in (a) for all three considered strains.

4.2. Shear test

In this example, we investigate the model behavior under cyclic loads, as outlined in Table 4. Two generalized strains are invoked to define the equilibrium part of the energy G_{iso}^∞ , and their parameters are $m_1 = 2.0$, $n_1 = 0.0$ and $m_2 = n_2 = 0.0$, respectively. A single set of parameters, denoted by m and n , is utilized to define the two strains in the configurational free energy γ^1 . The block is clamped at the bottom and subjected to a sinusoidal displacement loading $G_2(t) = U_0 \sin(\omega t)$ along the X_2 -direction on the top surface. The frequency is fixed to be $\omega = 0.3$. Stress-free boundary conditions are applied on the rest boundary surfaces. The shear

Table 6

Bearing test: problem setting. The dark color represents the steel layer with height being 0.003; the light color represents the rubber layer with height being 0.025. (For interpretation of the references to color, the reader is referred to the web version of this article.)

Rubber layers:

$$\begin{aligned}\rho_0 &= 1.0 \times 10^3, \\ N &= 2, M = 2, \\ G_{\text{iso}}^\infty &= \mu_1^\infty |\tilde{E}_1^\infty|^2 + \mu_2^\infty |\tilde{E}_2^\infty|^2, \\ \mu_1^\infty &= 1.75 \times 10^5, \quad \mu_2^\infty = 0.35 \times 10^5, \\ Y^1 &= \mu^1 |\tilde{E}^1 - E^{v1}|^2, \\ Y^2 &= \mu^2 |\tilde{E}^2 - E^{v2}|^2, \\ \tau^1 &= 17.5, \quad \tau^2 = 17.5, \\ \mu^1 &= 0.536 \times 10^4, \quad \mu^2 = 5.3064 \times 10^5.\end{aligned}$$

Steel layers:

$$\begin{aligned}G^{\text{st}} &= \mu^{\text{st}} |\mathbf{E}^{\text{st}}|^2, \\ \mu^{\text{st}} &= 8.077 \times 10^{10}, \\ m^{\text{st}} &= 2.0, n^{\text{st}} = 2.0.\end{aligned}$$

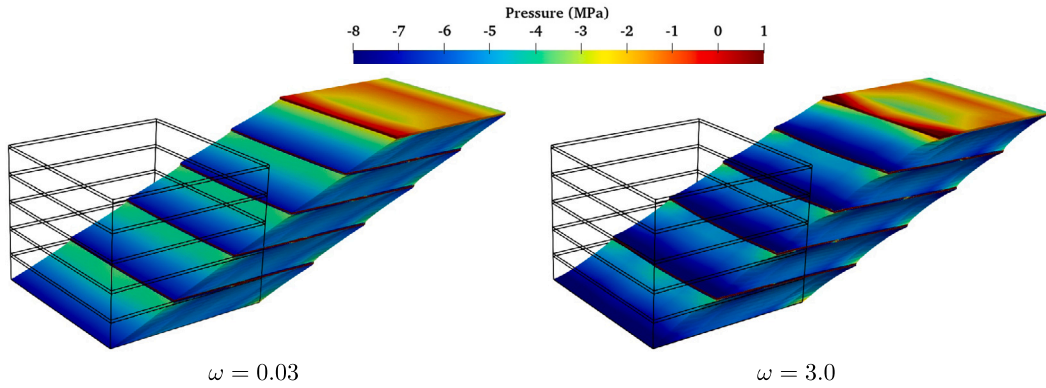
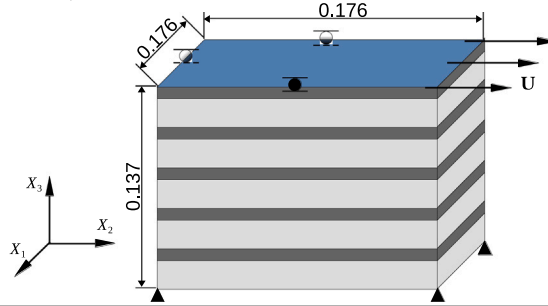


Fig. 10. Bearing test: the pressure distributions of the bearing at the maximum shear state with two strain rates.

strain on the top surface is given by $G_2(t)/L_0$ here, with $L_0 = 0.1$ being the edge length of the cube. We consider two cases with the maximum shear strains U_0/L_0 equal to 1% and 100%, respectively. Six loading cycles are performed in each simulation to achieve the dynamic equilibrium state [16, p. 3472]. The time step size Δt_n is fixed at 0.1. The spatial discretization is identical to that of the previous example.

We monitor the total traction on the top surface and plot its component in the X_2 -direction against the shear strain $G_2(t)/L_0$ in Fig. 9. The results illustrate the hysteresis loops corresponding to the two different loading parameters of U_0/L_0 . In particular, we consider the strains $\tilde{E}^1$ and E^{v1} parameterized by $(m, n) = (0, 0)$, $(1, 0)$, and $(2, 0)$. When $U_0/L_0 = 100\%$, the peak value of the force increases from 6000 (Fig. 9(b)) to 10000 (Fig. 9(d)) as m increases from 0 to 2. In contrast, when the shear is small (i.e., $U_0/L_0 = 1\%$), the influence of the parameter m on the force is negligible (Fig. 9(a)). We recall that the parameter m enters into the stress-strain relation as the power of the stretches, which may explain the aforesaid phenomena. Additionally, with the increase of the maximum shear strain, the strain rate gets stronger due to the fixed frequency ω , and this drives the material to behave more like a hyperelastic material. This observation is consistent with previous studies on the cyclic loading test of a viscoelastic body [16]. In Table 5, we observe quadratic convergence rates in the Newton-Raphson iterative procedure for all three considered strains.

4.3. Bearing

In the final example, we explore the shearing behavior of a multilayered rubber-steel structure, which is used for seismic isolation. The problem setting is summarized in Table 6. The steel layers are modeled by the generalized Hill's hyperelasticity model based on $\tilde{E}^{\text{st}}$, whose parameters are $m^{\text{st}} = 2.0$ and $n^{\text{st}} = 2.0$. As for the rubber layers, both the equilibrium and non-equilibrium parts are characterized by two quadratic energy terms. The generalized strains $\tilde{E}_1^\infty$, $\tilde{E}^1$, and E^{v1} are defined by the same scale function with parameters $m_1 = 2.0$ and $n_1 = 0.0$; the generalized strains $\tilde{E}_2^\infty$, $\tilde{E}^2$, and E^{v2} are defined by the same scale function with parameters $m_2 = 0.0$ and $n_2 = 0.0$.

The bearing is fixed on the bottom surface and subjected to a sinusoidal displacement loading along the X_2 -direction on the top surface with $G_2(t) = 0.35\sin(\omega t)$. Stress-free boundary conditions are applied on the remaining boundary surfaces. The geometry of the bearing is represented by multi-patch NURBS, with each layer constructed by a single patch. The presented results are based

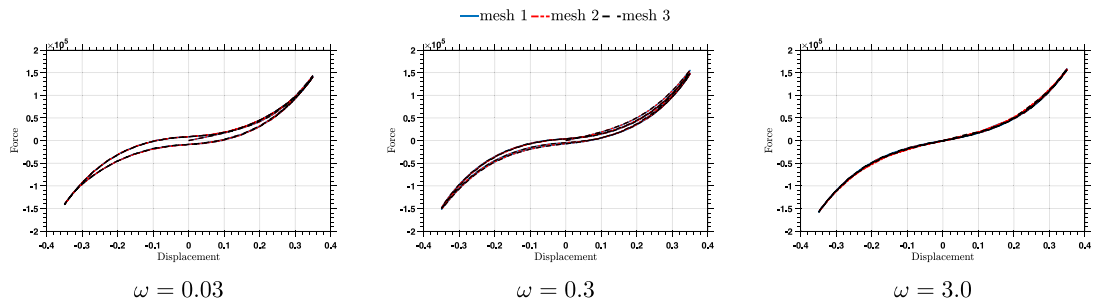


Fig. 11. Bearing test: the force–displacement curve for different strain rates.

on a spatial mesh of $3 \times 3 \times 3$ elements for each patch with $p = 2$. The results have been verified by a mesh independence study. The investigated frequencies ω are 0.03, 0.3, and 3.0, corresponding to time step sizes of 0.1, 0.01, and 0.0001, respectively. Each simulation is conducted for two cycles.

Fig. 10 depicts the pressure distributions when the multilayered structure reaches the maximum strain. As shown in the figure, higher values of ω results in more intense pressure distributions. Fig. 11 illustrates the total force against the displacement in the X_2 -direction on the top boundary surface. The curves obtained by three distinct meshes are plotted, demonstrating mesh independence. We observe that increasing the frequency ω leads to a slight increase of the peak value of the hysteresis curve, consistent with Fig. 10. Higher frequencies also cause the material to behave more like a hyperelastic material. These observations, again, align well with the prior studies on viscoelastic hysteresis [1,16].

5. Conclusion

In this work, we present a viscoelasticity model that extends the finite deformation linear viscoelastic model [1,3,10] to the nonlinear regime. Our approach stems from an examination of the relaxation property of the non-equilibrium stress. It is found to be intricately linked to the kinematic assumption akin to the Green–Naghdi type developed in elastoplasticity. This inspires a set of kinematic assumptions tailored for viscoelasticity, from which a nonlinear theory is systematically constructed. To maintain consistency with the existing models, we adopt the hyperelasticity of Hill’s class into our theory. We also survey recent advancements in the generalized strain concept, which offers ample opportunities in model design. Moreover, Hill’s hyperelasticity is generalized by combining multiple quadratic strain energy terms, and the improvement of this generalization is demonstrated through fitting experimental data. We constructed the theory based on the Helmholtz and Gibbs free energies, which are applicable to the pure displacement and mixed formulations, respectively. Within this framework, the nonlinear extension is achieved through generalized strains, and the previous linear model [1] gets recovered when the strain is instantiated to be of the Green–Lagrange type. This modeling framework differs from the theory based on the intermediate configuration in a fundamental manner. Since the internal state variable is a Lagrangian tensor belonging to $\text{Sym}(3)_+$, there is no necessity to make assumptions regarding the rotation of the intermediate configuration. This feature can be particularly beneficial in treating anisotropy as one does not need to model the evolution of the symmetry groups. The benefit of the Green–Naghdi assumption is also manifested in guaranteeing the satisfaction of the relaxation property of the non-equilibrium stress. This is a rigorously established property for models built upon this assumption.

The numerical implementation of the model is designed in a modular approach, encompassing the geometric preprocessing, constitutive evaluation, and geometric postprocessing. These steps are facilitated by the projection tensors induced from the generalized strains. Due to the nonlinearity, the evolution equation is integrated by the mid-point rule, and the internal state variables are determined locally at each quadrature point through a local Newton–Raphson iteration. Numerical examples, including creep and cyclic shear, are provided to illustrate the effectiveness of the model as well as the performance of the proposed algorithm. We also considered a bearing example to showcase the practical applicability of the proposed model.

One limitation of our study is that only one particular strain family is considered in the numerical examples. Based on the survey of the generalized strains, there are abundant possibilities for model design. It is essential to explore their mathematical validity, capability in characterizing real materials, and related numerical aspects. Our future work will also focus on developing a nonlinear anisotropic viscoelasticity model. It is also imperative to consider integration methods that preserve critical physical features, following the strategies developed in [103]. This will be useful for the evaluating the long-term performance of viscoelastic devices and providing insights into their nonlinear behaviors.

CRediT authorship contribution statement

Ju Liu: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Jiashen Guan:** Writing – original draft, Software, Methodology. **Chongran Zhao:** Validation, Software, Methodology. **Jiawei Luo:** Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Derivation of $\mathbb{Q}$

We start by deriving an explicit formula for the derivative of $\mathbf{M}_a$ with respect to $\mathbf{C}$. According to (2.2), the eigenvalue problem for $\mathbf{C}$ reads as

$$\mathbf{C} \mathbf{N}_a = \lambda_a^2 \mathbf{N}_a. \quad (\text{A.1})$$

Taking derivative with respect to $\mathbf{C}$ at both sides of (A.1) leads to

$$\frac{\partial (\mathbf{C} \mathbf{N}_a)}{\partial \mathbf{C}} = \frac{\partial (\lambda_a^2 \mathbf{N}_a)}{\partial \mathbf{C}},$$

which can be expanded as

$$\mathbf{N}_a \cdot \frac{\partial \mathbf{C}^T}{\partial \mathbf{C}} + \mathbf{C} \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \mathbf{N}_a \otimes \frac{\partial \lambda_a^2}{\partial \mathbf{C}} + \lambda_a^2 \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}}.$$

Performing the dot product of the above with the eigenvector $\mathbf{N}_b$, with $b \neq a$, results in

$$(\mathbf{N}_b \otimes \mathbf{N}_a) : \frac{\partial \mathbf{C}}{\partial \mathbf{C}} + \mathbf{N}_b \cdot \mathbf{C} \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = (\mathbf{N}_b \cdot \mathbf{N}_a) \frac{\partial \lambda_a^2}{\partial \mathbf{C}} + \lambda_a^2 \mathbf{N}_b \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \lambda_a^2 \mathbf{N}_b \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}}. \quad (\text{A.2})$$

The second term on the left-hand side can be rewritten as

$$\mathbf{N}_b \cdot \mathbf{C} \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = (\mathbf{C}^T \mathbf{N}_b) \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \lambda_b^2 \mathbf{N}_b \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}}, \quad (\text{A.3})$$

due to the symmetry of $\mathbf{C}$. With that, the relation (A.2) can be reorganized as

$$(\lambda_a^2 - \lambda_b^2) \mathbf{N}_b \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = (\mathbf{N}_b \otimes \mathbf{N}_a) : \frac{\partial \mathbf{C}}{\partial \mathbf{C}} = \frac{1}{2} (\mathbf{N}_a \otimes \mathbf{N}_b + \mathbf{N}_b \otimes \mathbf{N}_a), \quad (\text{A.4})$$

which further leads to

$$\mathbf{N}_b \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \frac{1}{2(\lambda_a^2 - \lambda_b^2)} (\mathbf{N}_a \otimes \mathbf{N}_b + \mathbf{N}_b \otimes \mathbf{N}_a).$$

From the above, the derivative of the eigenvector $\mathbf{N}_a$ with respect to $\mathbf{C}$ can be represented as

$$\frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \frac{1}{2} \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \mathbf{N}_b \otimes (\mathbf{N}_b \otimes \mathbf{N}_a + \mathbf{N}_a \otimes \mathbf{N}_b), \quad (\text{A.5})$$

with

$$\mathbf{N}_a \cdot \frac{\partial \mathbf{N}_a}{\partial \mathbf{C}} = \frac{1}{2} \frac{\partial (\mathbf{N}_a \cdot \mathbf{N}_a)}{\partial \mathbf{C}} = \mathbb{0}.$$

Employing the expression derived from (A.5), the derivative of $\mathbf{M}_a$ with respect to $\mathbf{C}$ can be written as

$$\begin{aligned} \frac{\partial \mathbf{M}_a}{\partial \mathbf{C}} &= \frac{\partial (\mathbf{N}_a \otimes \mathbf{N}_a)}{\partial \mathbf{C}} \\ &= \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \frac{1}{2} [\mathbf{N}_b \otimes \mathbf{N}_a \otimes (\mathbf{N}_b \otimes \mathbf{N}_a + \mathbf{N}_a \otimes \mathbf{N}_b) + \mathbf{N}_a \otimes \mathbf{N}_b \otimes (\mathbf{N}_b \otimes \mathbf{N}_a + \mathbf{N}_a \otimes \mathbf{N}_b)] \\ &= \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b). \end{aligned} \quad (\text{A.6})$$

With (A.6), we may obtain an explicit formula for $\mathbb{Q}$ as

$$\begin{aligned}\mathbb{Q} &= 2 \frac{\partial \mathbf{E}}{\partial \mathbf{C}} = \sum_{a=1}^3 \left(2 \frac{\partial E}{\partial (\lambda_a^2)} (\lambda_a) \mathbf{M}_a \otimes \frac{\partial \lambda_a^2}{\partial \mathbf{C}} \right) + \sum_{a=1}^3 2E(\lambda_a) \frac{\partial \mathbf{M}_a}{\partial \mathbf{C}} \\ &= \sum_{a=1}^3 d_a \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \vartheta_{ab} \mathbf{M}_a \odot \mathbf{M}_b,\end{aligned}$$

in which we invoked the relation [65, p. 91]

$$\frac{\partial \lambda_a^2}{\partial \mathbf{C}} = \mathbf{M}_a, \quad (\text{A.7})$$

and the terms d_a and ϑ_{ab} are given by (2.9).

Appendix B. Derivation of $\mathbb{L}$

We start the derivation of the explicit formula for $\mathbb{L}$ by considering the second derivative of $\mathbf{M}_a$ with respect to $\mathbf{C}$. Following (A.6), we have

$$\frac{\partial^2 \mathbf{M}_a}{\partial \mathbf{C} \partial \mathbf{C}} = \sum_{b \neq a}^3 (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b) \otimes \frac{\partial}{\partial \mathbf{C}} \left(\frac{1}{\lambda_a^2 - \lambda_b^2} \right) + \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b). \quad (\text{B.1})$$

The first term on the right-hand side can be written as

$$\begin{aligned}& \sum_{b \neq a}^3 (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b) \otimes \frac{\partial}{\partial \mathbf{C}} \left(\frac{1}{\lambda_a^2 - \lambda_b^2} \right) \\ &= \sum_{b \neq a}^3 \frac{1}{(\lambda_a^2 - \lambda_b^2)^2} (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b) \otimes \left(\frac{\partial \lambda_b^2}{\partial \mathbf{C}} - \frac{\partial \lambda_a^2}{\partial \mathbf{C}} \right) \\ &= \sum_{b \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)^2} [4(\mathbf{M}_b \odot \mathbf{M}_a) \otimes \mathbf{M}_b + 4(\mathbf{M}_a \odot \mathbf{M}_b) \otimes \mathbf{M}_b - 4(\mathbf{M}_b \odot \mathbf{M}_a) \otimes \mathbf{M}_a - 4(\mathbf{M}_a \odot \mathbf{M}_b) \otimes \mathbf{M}_a] \\ &= \sum_{b \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)^2} (\mathbb{H}_{abb} - \mathbb{H}_{baa}).\end{aligned} \quad (\text{B.2})$$

In the last equality of the above, we have invoked the following identities

$$\begin{aligned}\mathbb{H}_{abb} &= 4(\mathbf{M}_b \odot \mathbf{M}_a) \otimes \mathbf{M}_b + 4(\mathbf{M}_a \odot \mathbf{M}_b) \otimes \mathbf{M}_b, \\ \mathbb{H}_{baa} &= 4(\mathbf{M}_b \odot \mathbf{M}_a) \otimes \mathbf{M}_a + 4(\mathbf{M}_a \odot \mathbf{M}_b) \otimes \mathbf{M}_a.\end{aligned} \quad (\text{B.3})$$

due to the definition of $\mathbb{H}_{abc}$ given in (2.15). Next, we consider the derivative of $\mathbf{M}_a \odot \mathbf{M}_b$ with respect to $\mathbf{C}$. Through exploiting (A.5), it can be explicitly represented as

$$\begin{aligned}& \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_a \odot \mathbf{M}_b) \\ &= \sum_{b \neq a}^3 \frac{1}{(\lambda_a^2 - \lambda_b^2)} \left\{ \sum_{c \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_c^2)} [N_c \otimes N_b \otimes N_a \otimes N_b \otimes (N_c \otimes N_a + N_a \otimes N_c)] \right. \\ &\quad + \sum_{c \neq b}^3 \frac{1}{4(\lambda_b^2 - \lambda_c^2)} [N_a \otimes N_c \otimes N_a \otimes N_b \otimes (N_c \otimes N_b + N_b \otimes N_c)] \\ &\quad + \sum_{c \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_c^2)} [N_a \otimes N_b \otimes N_c \otimes N_b \otimes (N_c \otimes N_a + N_a \otimes N_c)] \\ &\quad + \sum_{c \neq b}^3 \frac{1}{4(\lambda_b^2 - \lambda_c^2)} [N_a \otimes N_b \otimes N_a \otimes N_c \otimes (N_c \otimes N_b + N_b \otimes N_c)] \\ &\quad + \sum_{c \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_c^2)} [N_c \otimes N_b \otimes N_b \otimes N_a \otimes (N_c \otimes N_a + N_a \otimes N_c)] \\ &\quad + \sum_{c \neq b}^3 \frac{1}{4(\lambda_b^2 - \lambda_c^2)} [N_a \otimes N_c \otimes N_b \otimes N_a \otimes (N_c \otimes N_b + N_b \otimes N_c)] \\ &\quad \left. + \sum_{c \neq b}^3 \frac{1}{4(\lambda_b^2 - \lambda_c^2)} [N_a \otimes N_b \otimes N_c \otimes N_a \otimes (N_c \otimes N_b + N_b \otimes N_c)] \right\}\end{aligned}$$

$$+ \sum_{c \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_c^2)} [N_a \otimes N_b \otimes N_b \otimes N_c \otimes (N_c \otimes N_a + N_a \otimes N_c)] \Big\}.$$

The above can be reorganized as

$$\begin{aligned} & \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_a \odot \mathbf{M}_b) \\ = & \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} [N_c \otimes N_b \otimes N_a \otimes N_b \otimes (N_c \otimes N_a + N_a \otimes N_c) \\ & + N_a \otimes N_b \otimes N_c \otimes N_b \otimes (N_c \otimes N_a + N_a \otimes N_c) \\ & + N_c \otimes N_b \otimes N_b \otimes N_a \otimes (N_c \otimes N_a + N_a \otimes N_c) \\ & + N_a \otimes N_b \otimes N_b \otimes N_c \otimes (N_c \otimes N_a + N_a \otimes N_c)] \\ & + \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_b^2 - \lambda_c^2)} [N_a \otimes N_c \otimes N_a \otimes N_b \otimes (N_c \otimes N_b + N_b \otimes N_c) \\ & + N_a \otimes N_b \otimes N_a \otimes N_c \otimes (N_c \otimes N_b + N_b \otimes N_c) \\ & + N_a \otimes N_c \otimes N_b \otimes N_a \otimes (N_c \otimes N_b + N_b \otimes N_c) \\ & + N_a \otimes N_b \otimes N_c \otimes N_a \otimes (N_c \otimes N_b + N_b \otimes N_c)] \\ & + \sum_{b \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)^2} [N_b \otimes N_b \otimes N_a \otimes N_b \otimes (N_b \otimes N_a + N_a \otimes N_b) \\ & + N_a \otimes N_b \otimes N_b \otimes N_b \otimes (N_b \otimes N_a + N_a \otimes N_b) \\ & + N_b \otimes N_b \otimes N_b \otimes N_a \otimes (N_b \otimes N_a + N_a \otimes N_b) \\ & + N_a \otimes N_b \otimes N_b \otimes N_b \otimes (N_b \otimes N_a + N_a \otimes N_b)] \\ & - [N_a \otimes N_a \otimes N_a \otimes N_b \otimes (N_a \otimes N_b + N_b \otimes N_a) \\ & + N_a \otimes N_b \otimes N_a \otimes N_a \otimes (N_a \otimes N_b + N_b \otimes N_a) \\ & + N_a \otimes N_a \otimes N_b \otimes N_a \otimes (N_a \otimes N_b + N_b \otimes N_a) \\ & + N_a \otimes N_b \otimes N_a \otimes N_a \otimes (N_a \otimes N_b + N_b \otimes N_a)] . \end{aligned} \quad (\text{B.4})$$

With the definition of $\mathbb{H}$ given in (2.15) and (B.4), the second term on the right-hand side of (B.1) can be represented as

$$\begin{aligned} & \sum_{b \neq a}^3 \frac{1}{\lambda_a^2 - \lambda_b^2} \frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_b \odot \mathbf{M}_a + \mathbf{M}_a \odot \mathbf{M}_b) \\ = & \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} [\mathbb{H}_{bca} + \mathbb{H}_{bac}] + \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_b^2 - \lambda_c^2)} [\mathbb{H}_{abc} + \mathbb{H}_{acb}] \\ & + \sum_{b \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)^2} [(\mathbb{H}_{bba} + \mathbb{H}_{bab}) - (\mathbb{H}_{aab} + \mathbb{H}_{aba})] . \end{aligned} \quad (\text{B.5})$$

Combining (B.2) and (B.5) leads to

$$\begin{aligned} \frac{\partial^2 \mathbf{M}_a}{\partial \mathbf{C} \partial \mathbf{C}} = & \sum_{b \neq a}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)^2} [(\mathbb{H}_{bba} + \mathbb{H}_{bab} + \mathbb{H}_{abb}) - (\mathbb{H}_{aab} + \mathbb{H}_{aba} + \mathbb{H}_{baa})] \\ & + \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} [\mathbb{H}_{bca} + \mathbb{H}_{bac} + \mathbb{H}_{abc}] , \end{aligned} \quad (\text{B.6})$$

in which we have invoked the following identity

$$\sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_b^2 - \lambda_c^2)} [\mathbb{H}_{abc} + \mathbb{H}_{acb}] = \sum_{b \neq a}^3 \sum_{\substack{c \neq a \\ c \neq b}}^3 \frac{1}{4(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} \mathbb{H}_{abc}.$$

According to (B.6) and (A.7), we may have $\mathbb{L}$ as

$$\mathbb{L} = 4 \frac{\partial^2 \mathbf{E}}{\partial \mathbf{C} \partial \mathbf{C}} = 4 \frac{\partial}{\partial \mathbf{C}} \left[\sum_{a=1}^3 \left(\frac{\partial E}{\partial (\lambda^2)} (\lambda_a) \mathbf{M}_a \otimes \mathbf{M}_a \right) + \sum_{a=1}^3 E(\lambda_a) \frac{\partial \mathbf{M}_a}{\partial \mathbf{C}} \right]$$

$$\begin{aligned}
&= \sum_{a=1}^3 4 \frac{\partial^2 E}{\partial(\lambda^2) \partial(\lambda^2)} (\lambda_a) \mathbf{M}_a \otimes \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 4 \frac{\partial E}{\partial(\lambda^2)} (\lambda_a) \frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_a \otimes \mathbf{M}_a) \\
&\quad + \sum_{a=1}^3 4 \frac{\partial E}{\partial(\lambda^2)} (\lambda_a) \frac{\partial \mathbf{M}_a}{\partial \mathbf{C}} \otimes \mathbf{M}_a + \sum_{a=1}^3 4 E(\lambda_a) \frac{\partial^2 \mathbf{M}_a}{\partial \mathbf{C} \partial \mathbf{C}}.
\end{aligned}$$

Combining the results of (2.9)₁, (2.12), (2.15), (A.5), (A.6), (B.3) and (B.6), $\mathbb{L}$ may be represented as

$$\begin{aligned}
\mathbb{L} &= \sum_{a=1}^3 f_a \mathbf{M}_a \otimes \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \frac{d_a}{2(\lambda_a^2 - \lambda_b^2)} (\mathbb{H}_{aba} + \mathbb{H}_{aab} + \mathbb{H}_{baa}) \\
&\quad + \sum_{b \neq a}^3 \frac{E(\lambda_a)}{(\lambda_a^2 - \lambda_b^2)^2} [(\mathbb{H}_{bba} + \mathbb{H}_{bab} + \mathbb{H}_{abb}) - (\mathbb{H}_{aab} + \mathbb{H}_{aba} + \mathbb{H}_{baa})] \\
&\quad + \sum_{b \neq a}^3 \sum_{c \neq a}^3 \frac{E(\lambda_a)}{(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} [\mathbb{H}_{bca} + \mathbb{H}_{bac} + \mathbb{H}_{abc}].
\end{aligned} \tag{B.7}$$

In the above, we have utilized the following,

$$\frac{\partial}{\partial \mathbf{C}} (\mathbf{M}_a \otimes \mathbf{M}_a) = \frac{1}{4} (\mathbb{H}_{aba} + \mathbb{H}_{aab}), \quad \frac{\partial \mathbf{M}_a}{\partial \mathbf{C}} = \frac{1}{4} \mathbb{H}_{baa}.$$

With (B.7), we may obtain an explicit formula for $\mathbb{L}$ as

$$\mathbb{L} = \sum_{a=1}^3 f_a \mathbf{M}_a \otimes \mathbf{M}_a \otimes \mathbf{M}_a + \sum_{a=1}^3 \sum_{b \neq a}^3 \xi_{ab} (\mathbb{H}_{abb} + \mathbb{H}_{bab} + \mathbb{H}_{bba}) + \sum_{a=1}^3 \sum_{b \neq a}^3 \sum_{c \neq a}^3 \eta_{abc} \mathbb{H}_{abc},$$

in which we invoked the following identities,

$$\sum_{a=1}^3 \sum_{b \neq a}^3 \frac{d_a}{2(\lambda_a^2 - \lambda_b^2)} (\mathbb{H}_{aba} + \mathbb{H}_{aab} + \mathbb{H}_{baa}) = - \sum_{a=1}^3 \sum_{b \neq a}^3 \frac{d_b}{2(\lambda_a^2 - \lambda_b^2)} (\mathbb{H}_{abb} + \mathbb{H}_{bab} + \mathbb{H}_{bba}), \tag{B.8}$$

$$\sum_{b \neq a}^3 \frac{E(\lambda_a)}{(\lambda_a^2 - \lambda_b^2)^2} [(\mathbb{H}_{bba} + \mathbb{H}_{bab} + \mathbb{H}_{abb}) - (\mathbb{H}_{aab} + \mathbb{H}_{aba} + \mathbb{H}_{baa})] = \sum_{a=1}^3 \sum_{b \neq a}^3 \frac{\vartheta_{ab}}{2(\lambda_a^2 - \lambda_b^2)} (\mathbb{H}_{abb} + \mathbb{H}_{bab} + \mathbb{H}_{bba}), \tag{B.9}$$

$$\begin{aligned}
&\sum_{b \neq a}^3 \sum_{c \neq a}^3 \frac{E(\lambda_a)}{(\lambda_a^2 - \lambda_b^2)(\lambda_a^2 - \lambda_c^2)} [\mathbb{H}_{bca} + \mathbb{H}_{bac} + \mathbb{H}_{abc}] \\
&= \sum_{a=1}^3 \sum_{b \neq a}^3 \sum_{c \neq a}^3 \left[\sum_{d=1}^3 \sum_{e \neq d}^3 \sum_{f \neq e}^3 \frac{E(\lambda_d)}{2(\lambda_d^2 - \lambda_e^2)(\lambda_d^2 - \lambda_f^2)} \right] \mathbb{H}_{abc},
\end{aligned} \tag{B.10}$$

and the terms ξ_{ab} and η are given by (2.13) and (2.14), respectively.

Appendix C. A discourse on the connection between the two kinematic assumptions

In this section, we briefly explore the relationship between the multiplicative and additive decompositions. In the assumption of the multiplicative decomposition, the deformation gradient is decomposed as $\mathbf{F} = \mathbf{F}^e \mathbf{F}^v$. This leads to the elastic and viscous deformation tensors as

$$\mathbf{C}^e := \mathbf{F}^{eT} \mathbf{F}^e = \mathbf{F}^{v-T} \mathbf{C} \mathbf{F}^{v-1} \quad \text{and} \quad \mathbf{C}^v := \mathbf{F}^{vT} \mathbf{F}^v.$$

Similar to the definition (2.2), the two enjoy the following spectral decompositions

$$\mathbf{C}^e := \sum_{a=1}^3 \lambda_a^{e2} \hat{\mathbf{N}}_a \otimes \hat{\mathbf{N}}_a \quad \text{and} \quad \mathbf{C}^v := \sum_{a=1}^3 \lambda_a^{v2} \mathbf{N}_a \otimes \mathbf{N}_a,$$

in which, $\{\lambda_a^e\}$, $\{\lambda_a^v\}$, and $\{\hat{\mathbf{N}}_a\}$, for $a = 1, 2, 3$, represent the elastic principal stretches, viscous principal stretches, and principle directions related to the intermediate configuration, respectively. Following (2.3), we define two generalized strains $\hat{\mathbf{E}}^e$ and $\hat{\mathbf{E}}^v$ as

$$\hat{\mathbf{E}}^e := \sum_{a=1}^3 E(\lambda_a^e) \hat{\mathbf{N}}_a \otimes \hat{\mathbf{N}}_a \quad \text{and} \quad \hat{\mathbf{E}}^v := \sum_{a=1}^3 E^v(\lambda_a^v) \mathbf{N}_a \otimes \mathbf{N}_a.$$

In the above, we use hats to distinguish them from their counterparts in the additive theory. The first result is related to the equivalence of the strain invariants when the strain is of the Green–Lagrange type.

Proposition 5. If we interpret the internal state variable Γ as C^v , the basic invariants of $\hat{E}_{SH}^{e(2)}$ and $E_{SH}^{e(2)}\Gamma^{-1}$ coincide, i.e.

$$J_i(\hat{E}_{SH}^{e(2)}) = J_i(E_{SH}^{e(2)}\Gamma^{-1}), \quad \text{for } i = 1, 2, 3.$$

In the above, the basic invariants are defined as the traces of tensor powers, i.e., $J_i(\cdot) := \text{tr}((\cdot)^i)$, where $(\cdot)$ represents a rank-two tensor.

Proof. The first basic invariant of $\hat{E}_{SH}^{e(2)}$ is

$$J_1(\hat{E}_{SH}^{e(2)}) = \text{tr}(\hat{E}_{SH}^{e(2)}) = \frac{1}{2}(\text{tr}(F^{v-T}CF^{v-1}) - 3) = \frac{1}{2}(\text{tr}(CC^{v-1}) - 3).$$

With the interpretation of Γ as C^v , we have

$$E_{eSH}^{(2)}C^{v-1} = \frac{1}{2}(CC^{v-1} - \Gamma C^{v-1}) = \frac{1}{2}(CC^{v-1} - I).$$

Subsequently, we have

$$J_1(E_{eSH}^{(2)}C^{v-1}) = \text{tr}(E_{eSH}^{(2)}C^{v-1}) = \frac{1}{2}(\text{tr}(CC^{v-1}) - 3) = J_1(\hat{E}_{SH}^{e(2)}).$$

From the derivation above, we establish the identity $\text{tr}(CC^{v-1}) = \text{tr}(C^e)$. After straightforward computations, we may further obtain

$$\begin{aligned} \text{tr}((CC^{v-1})^2) &= (CC^{v-1}) : (C^{v-1}C) \\ &= (F^{v-T}CF^{v-1}) : (F^{v-T}CF^{v-1}) \\ &= C^e : C^e \\ &= \text{tr}(C^{e2}) \end{aligned}$$

and

$$\begin{aligned} \text{tr}((CC^{v-1})^3) &= (CC^{v-1}CC^{v-1}) : (C^{v-1}C) \\ &= (F^{v-T}CC^{v-1}CF^{v-1}) : (F^{v-T}CF^{v-1}) \\ &= C^{e2} : C^e \\ &= \text{tr}(C^{e3}). \end{aligned}$$

Regarding the second and third basic invariants, we have

$$J_2(\hat{E}_{SH}^{e(2)}) = \text{tr}\left(\left(\hat{E}_{SH}^{e(2)}\right)^2\right) = \frac{1}{4}(\text{tr}(C^{e2}) - 2\text{tr}C^e + 3) = \frac{1}{4}\left(\text{tr}((CC^{v-1})^2) - 2\text{tr}(CC^{v-1}) + 3\right),$$

and

$$\begin{aligned} J_3(\hat{E}_{SH}^{e(2)}) &= \text{tr}\left(\left(\hat{E}_{SH}^{e(2)}\right)^3\right) = \frac{1}{8}(\text{tr}(C^{e3}) - 3\text{tr}(C^{e2}) + 3\text{tr}C^e - 3) \\ &= \frac{1}{8}(\text{tr}((CC^{v-1})^3) - 3\text{tr}((CC^{v-1})^2) + 3\text{tr}(CC^{v-1}) - 3). \end{aligned}$$

With the following relations

$$\begin{aligned} CC^{v-1} &= 2E_{SH}^{(2)}C^{v-1} + I, \\ (CC^{v-1})^2 &= 4(E_{SH}^{(2)}C^{v-1})^2 + 4E_{SH}^{(2)}C^{v-1} + I, \\ (CC^{v-1})^3 &= 8(E_{SH}^{(2)}C^{v-1})^3 + 12(E_{SH}^{(2)}C^{v-1})^2 + 6E_{SH}^{(2)}C^{v-1} + I, \end{aligned}$$

we obtain the following,

$$\begin{aligned} J_2(\hat{E}_{SH}^{e(2)}) &= \text{tr}\left(\left(E_{SH}^{(2)}C^{v-1}\right)^2\right) = J_2(E_{SH}^{(2)}C^{v-1}), \\ J_3(\hat{E}_{SH}^{e(2)}) &= \text{tr}\left(\left(E_{SH}^{(2)}C^{v-1}\right)^3\right) = J_3(E_{SH}^{(2)}C^{v-1}), \end{aligned}$$

which completes the proof. $\square$

In the second result, we consider the Hencky strain and assume the deformation gradient F^v is a pure stretch tensor and coaxial with C . This assumption suggests that F^v commutes with C , which eventually leads to an additive split structure of the strain within the multiplicative decomposition framework.

Proposition 6. If F^v is a pure stretch tensor coaxial with C , we have the following additive split structure of strains

$$E_{SH}^{(0)} = \hat{E}_{SH}^{e(0)} + \hat{E}_{SH}^{v(0)}.$$

Proof. Given the multiplicative decomposition $F = F^e F^v$, the elastic Hencky strain is given by

$$\hat{E}_{SH}^{e(0)} = \frac{1}{2} \ln(C^e) = \frac{1}{2} \ln(F^{v-T} C F^{v-1}).$$

Since F^v is a pure stretch tensor coaxial with C , we have

$$\hat{E}_{SH}^{e(0)} = \frac{1}{2} \ln(C F^{v-T} F^{v-1}) = \frac{1}{2} \sum_{a=1}^3 \ln(\lambda_a^2 \lambda_a^{v-2}) N_a \otimes N_a = E_{SH}^{(0)} - \hat{E}_{SH}^{v(0)},$$

which completes the proof. $\square$

Appendix D. Glossary of terms

Symbol	Name or description	Place of definition or first occurrence
E	The generalized strain	(2.3)
$\tilde{E}$	The generalized strain associated with isochoric deformation	(2.4)
Q, L	The first and second derivatives of the strain E with respect to C	(2.8), (2.11)
$\tilde{Q}, \tilde{L}$	The first and second derivatives of the strain $\tilde{E}$ with respect to $\tilde{C}$	(2.16)
I, Q, E^v	The deformation-like internal state variable, the stress-like internal state variable, and the viscous strain	(2.24), (2.30), (2.25)
Q^v, L^v	The first and second derivatives of the strain E^v with respect to I	(2.26)
Ψ, Ψ^∞, Y	The Helmholtz free energy, its equilibrium part, and the configurational free energy	(2.28)
D	The internal dissipation	(2.29)
$\mathbb{V}$	The viscosity tensor	(2.31)
$S, S^\infty, S^{\text{neq}}$	The second Piola–Kirchhoff stress, its equilibrium part, and its non-equilibrium part	(2.31), (2.35) ₁ , (2.35) ₂
T^∞, T^{neq}	The stresses conjugate to the generalized strain E with respect to Ψ^∞ and Y , respectively	(2.35) ₃ , (2.35) ₄
$E^\alpha, E^{v\alpha}$	The generalized strains that define Y^α	Section 2.5.1
Ψ_β^∞	The β th quadratic term of Ψ^∞	(2.37) ₁
E_β^∞	The generalized strain used to define the quadratic energy Ψ_β^∞	(2.37)
Y^α	The configurational free energy associated with the α th relaxation process	(2.37) ₂
T_β^∞	The stress conjugate to E_β with respect to Ψ_β^∞	(2.39) ₁
T^α	The stress conjugate to E^α with respect to Y^α	(2.39) ₂
Q_β^∞	The first derivative of the strain E_β^∞ with respect to C	(2.40) ₁
$Q^\alpha, Q^{v\alpha}$	The derivative of E^α with respect to C and the derivative of $E^{v\alpha}$ with respect to I^α	(2.40) ₂ , (2.40) ₃
$I^\alpha, Q^\alpha, \mathbb{V}^\alpha$	The internal state variables and viscosity tensors associated with the α th relaxation processes	Section 2.4, (2.38) ₃ , (2.41) ₁
$\Psi_{\text{vol}}^\infty, \Psi_{\text{iso}}^\infty, \Psi_{\text{iso}}^\infty$	The volumetric part of Ψ , the isochoric part of Ψ , and the isochoric part of Ψ^∞	(2.42), (2.43)
$G, G_{\text{vol}}^\infty, G_{\text{iso}}^\infty, G_{\text{iso}}^\infty$	The Gibbs free energy, its volumetric part, its isochoric part, and the equilibrium part of G_{iso}^∞	(2.44), (2.45)
$\mu^\infty, \mu^{\text{neq}}$	The shear moduli for the equilibrium part and non-equilibrium part	(2.46)
$\tilde{S}_{\text{iso}}^\infty, \tilde{S}_{\text{iso}}^{\text{neq}}$	Fictitious stresses	(2.47) ₁ , (2.47) ₂
$\tilde{T}^\infty, \tilde{T}^{\text{neq}}$	The stresses conjugate to the generalized strain $\tilde{E}$ with respect to G_{iso}^∞ and Y , respectively	(2.51)
$G_{\text{iso}\beta}^\infty$	The β th quadratic term of G_{iso}^∞	(2.54) ₁
$\tilde{E}_\beta^\infty$	The generalized strains used to define the quadratic energy $G_{\text{iso}\beta}^\infty$	(2.54) ₁
$\mu_\beta^\infty, \mu^\alpha$	The shear moduli used in $G_{\text{iso}\beta}^\infty$ and Y^α	(2.54)
$\tilde{E}^\alpha$	The strain used in the definition of the quadratic energy Y^α	(2.54)
$\tilde{T}_\beta^\infty, \tilde{T}^\alpha$	The stress conjugate to $\tilde{E}_\beta^\infty$ with respect to $G_{\text{iso}\beta}^\infty$ and the stress conjugate to $\tilde{E}^\alpha$ with respect to Y^α , respectively	(2.56) ₁ , (2.56) ₂

$\tilde{Q}_\beta^\infty, \tilde{L}_\beta^\infty$	The first and second derivatives of $\tilde{E}_\beta^\infty$ with respect to $\tilde{C}$	(2.57) ₁ , (3.11)
$\tilde{Q}_\alpha^\infty, \tilde{L}_\alpha^\infty$	The first and second derivatives of $\tilde{E}_\alpha^\infty$ with respect to $\tilde{C}$	(2.57) ₂ , (3.14)
K^α	The derivative of Q^α with respect to Γ^α	(3.2)
J^α	The derivative of Q^α with respect to $\tilde{C}$	(3.3)
$L^{\vee\alpha}$	The second derivative of $E^{\vee\alpha}$ with respect to Γ^α	(3.4)
S^α	The rank-four tensor used in constitutive integration	(3.6) , (3.15)
$C_{iso}, C_{iso}^\infty, C_{iso}^{neq}$	The isochoric elasticity tensor, its equilibrium and non-equilibrium parts	(3.7)
$\tilde{C}_{iso}^\infty, \tilde{C}_{iso}^{neq}$	The fictitious elasticity tensors	(3.9) ₁ , (3.13)

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